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# From the Discrete Model to the Black-Scholes Formula

*Risk-Neutral Measure, Arbitrage,  
Complete Markets and Option Pricing*

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On October 16, 1973, Fischer Black and Myron Scholes published in the *Journal of Political Economy* an article entitled *The Pricing of Options and Corporate Liabilities*. A few weeks later, Robert Merton published in the *Bell Journal of Economics* an extension of this work. These two articles founded modern mathematical finance, in the sense that they formulated for the first time a rigorous answer to an apparently simple question: what is the fair price of an option?

The answer they provide is remarkable on two counts. Formally, it is expressed by a partial differential equation (the Black-Scholes equation) whose explicit solution gives, for a European *call* with underlying  $S$ , strike  $K$ , maturity  $T$ , risk-free rate  $r$  and volatility  $\sigma$ <sup>1</sup>:

$$C = S \phi(d) - K e^{-rT} \phi(d - \sigma\sqrt{T})$$

with  $d = \frac{\ln(S/K) + rT}{\sigma\sqrt{T}} + \frac{\sigma\sqrt{T}}{2}$ , where  $\phi$  is the cumulative distribution function of  $\mathcal{N}(0, 1)$ . Economically, this formula is even more surprising: the price of the option does not depend on the expected return of the stock that investors attribute to it. The price depends on observable parameters:  $\sigma$ ,  $r$ ,  $K$  and  $T$ , but not on the probability that the price rises or falls. As if the risk aversion of agents had vanished from the equation.

The central question of this project will be to understand why this formula is correct, what it truly states, and how it is rigorously derived.

This project draws predominantly from Chapters 1 and 2 of the reference work *Financial Markets in Continuous Time* by Rose-Anne Dana and Monique Jeanblanc (Springer Finance, corrected edition 2007), covering the following sections:

- Chapter 1, Sections 1.1 (*A Model with Two Dates and Two States of the World*) through 1.2.4 (*Incomplete Markets: the Arbitrage Interval*): the one- and multi-period model.
- Chapter 2, Sections 2.1 (*A Model with a Finite Horizon*) through 2.6.3 (*Approaching the Black-Scholes Model*): the dynamic discrete-time model and convergence towards Black-Scholes.

These sections form a coherent whole: they establish the theory of arbitrage pricing in the discrete framework, starting from the simplest model and culminating in the Black-Scholes formula.

A practical perspective not present in the reference work is provided by a complete numerical implementation in Python. Each theoretical result is translated into code, each identity is verified numerically, and the model is tested against real market data on a Crédit Agricole warrant.

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<sup>1</sup>All these notions will of course be defined and elaborated upon.

# Chapter 1

## The two-state binomial model

This first chapter constitutes the foundation of the entire dissertation. We place ourselves in the simplest possible setting, namely a financial market with two dates, two states of the world, and consisting of two assets: a risky asset and a risk-free asset. This will allow us to clearly bring out the fundamental concepts of the theory of financial option pricing: hedging portfolio, arbitrage, risk-neutral probability measure, and market completeness.

To introduce these notions, it is first necessary to describe the time framework and the various scenarios of this market. We thus consider a financial market comprising two dates: the initial date  $t = 0$  (today) and the terminal date  $t = 1$  (the maturity, for instance tomorrow or in 3 months). At date  $t = 1$ , the world may find itself in one of two states: an "up" state, corresponding to a rise in price, and a "down" state, corresponding to a fall. Of course, the outcome is unknown at  $t = 0$ . The market consists of two assets:

- A risky asset (a stock) worth  $S$  euros at  $t = 0$ . At  $t = 1$ , the stock will be worth  $S_h$  or  $S_b$  euros with  $0 < S_b < S_h$ .  $S_h$  corresponds to the up state and  $S_b$  to the down state.
- A risk-free asset (such as a savings account) yielding a rate  $r > 0$ . That is, one euro invested in your savings account is worth  $1 + r$  euros at  $t = 1$ , regardless of the state of the world, which justifies the designation "risk-free".

From these two basic assets, it is possible to construct and evaluate more complex financial products. We may then define the derivative product that is the subject of this dissertation: the option.

**Definition 1.1.** *A call option is a financial instrument: the buyer pays the seller, at  $t = 0$ , a sum  $q$  (the price) which gives them the right, but not the obligation, to purchase at  $t = 1$  the stock at a price  $K$  (exercise price) fixed at  $t = 0$ .*

*A put option is a financial instrument that gives its buyer the right to sell a stock at  $t = 1$  at a price  $K$  fixed at  $t = 0$ .*

Concretely, the idea is as follows: if  $S_1 > K$  (where  $S_1$  denotes the price of the asset at  $t = 1$ ), the holder of the call buys the stock at price  $K$  and immediately resells it, thereby realising a gain; otherwise, they do not buy and let the option expire, as it would make no sense to purchase a stock at a price above its market value. Conversely, for a put the holder will exercise the option if  $S_1 < K$ . In all cases, the seller of the option collects a

premium  $q$ , the price of the option, whose determination is precisely the subject of this dissertation.

It is then pertinent to observe that the price of a call (denoted  $C$ ) at  $t = 1$  can be written:

$$\begin{cases} C_h = \max(0, S_h - K) & \text{if up state} \\ C_b = \max(0, S_b - K) & \text{if down state} \end{cases}$$

In the same way, we obtain for a put (denoted  $P$ ):

$$\begin{cases} P_h = \max(0, K - S_h) & \text{if up state} \\ P_b = \max(0, K - S_b) & \text{if down state} \end{cases}$$

It is important to note that we are dealing here with **European** options, which can only be exercised at maturity. For instance, there exist American options, where it is possible to exercise the right at any date between the purchase and the maturity of the option. The valuation method is then different<sup>1</sup>.

We will now define what a *portfolio* is, an essential object for option pricing.

**Definition 1.2.** A *portfolio* is a pair  $(\alpha, \beta) \in \mathbb{R}^2$  where:

- $\alpha \in \mathbb{R}$  is the amount invested in the risk-free asset;
- $\beta \in \mathbb{R}$  is the number of units of stock held.

Its initial value (at  $t = 0$ ) is:

$$V_0 = \alpha + \beta S$$

Its terminal value (at  $t = 1$ ) depends on the state of the world:

$$V_1 = \begin{cases} \alpha(1 + r) + \beta S_h & \text{if up state} \\ \alpha(1 + r) + \beta S_b & \text{if down state} \end{cases}$$

Note that the quantities  $\alpha$  and  $\beta$  may be negative:  $\beta < 0$  corresponds to a short sale of the stock, and  $\alpha < 0$  to a borrowing.

The introduction of this mathematical tool makes it possible to formalise a method for valuing the option, whose principle rests on the replication of its future cash flows. The idea is to construct a portfolio that behaves like the option in every state of the world, and then to "travel back in time" (to  $t = 0$ ) in order to find its initial price. Such a portfolio is called a "*hedging portfolio*". Let us therefore seek the portfolio  $(\alpha^*, \beta^*)$  that replicates a European call. By writing the two replication conditions, namely that the replicating portfolio in the up state (resp. down state) must equal the price of the call in the up state (resp. down state), we obtain the linear system:

$$\begin{cases} \alpha(1 + r) + \beta S_h = C_h \\ \alpha(1 + r) + \beta S_b = C_b \end{cases}$$

Solving this system, since  $S_h \neq S_b$ , we obtain:

$$\alpha^* = \frac{1}{1 + r} \left( C_b - \frac{C_h - C_b}{S_h - S_b} S_b \right), \quad \beta^* = \frac{C_h - C_b}{S_h - S_b}$$

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<sup>1</sup>Cf. Chapter 4 for a detailed study of these options.

Thus, using the initial value of the replicating portfolio, the price  $q$  is given by:

$$q = \alpha^* + \beta^* S = \frac{1}{1+r} (\pi C_h + (1-\pi) C_b) \quad (1.1)$$

with:

$$\pi = \frac{1}{S_h - S_b} ((1+r)S - S_b) \quad (1.2)$$

The same construction yields the price of the put:

$$P = \frac{1}{1+r} (\pi P_h + (1-\pi) P_b)$$

From an economic standpoint, the amount  $q$  is the "fair price" of the option. The logic is as follows: the seller of the option collects  $q$  at  $t = 0$ , constructs the portfolio  $(\alpha^*, \beta^*)$ , and is thereby perfectly hedged (hence the name): whatever happens at  $t = 1$ , their portfolio produces exactly what they must pay the buyer (0 or  $S_h - K$  or  $S_b - K$ ). They bear no net risk. As for the buyer, they should not pay more than  $q$ , because if they did — say  $q = 10$  euros and the seller sells the option for 12 euros — the buyer would simply need to construct the replicating portfolio  $(\alpha^*, \beta^*)$  (for 10 euros) and invest the remaining 2 euros in the risk-free asset. The buyer would then obtain the same payoffs with an additional bonus from the remainder.

The quantity  $\pi$  defined in (1.2) plays a central role throughout the entire theory. When  $S_b < (1+r)S < S_h$ , we have  $\pi \in [0, 1]$ , and therefore naturally  $(1-\pi) \in [0, 1]$  as well. We may then define a probability measure as follows.

**Definition 1.3.** *The risk-neutral probability measure (or equivalent martingale measure<sup>2</sup>) is the measure  $\mathbb{Q}$  defined by*

$$\mathbb{Q}(S_1 = S_h) = \pi, \quad \mathbb{Q}(S_1 = S_b) = 1 - \pi$$

Its fundamental property is that the expectation of the discounted price of the stock equals its initial price:

$$\mathbb{E}^{\mathbb{Q}} \left[ \frac{S_1}{1+r} \right] = \frac{\pi S_h + (1-\pi) S_b}{1+r} = S. \quad (1.3)$$

The risky asset is therefore a *martingale* under  $\mathbb{Q}$  after discounting. This is precisely the property defining the risk-neutral measure in the general framework.

Let us note that we have *discounted* the price of the stock. Indeed, directly comparing an amount available today ( $S$ ) with a future cash flow ( $S_1$ ) is not possible, as the time value of money renders these sums non-comparable. In order to bring  $S_1$  back to its present equivalent value, we use discounting (via the factor  $\frac{1}{1+r}$ ). This makes it possible to isolate the true performance of the asset by eliminating the component attributable to risk-free interest.

The equality (1.3) can be rewritten as  $(1+r)S = \pi S_h + (1-\pi) S_b$ . That is, under  $\mathbb{Q}$ , the expected return of the stock equals the risk-free rate  $r$ . It is as though the investor were indifferent to risk, hence the name. It is essential to note that  $\mathbb{Q}$  **is not** the subjective probability modelling the event that the up state occurs; it is constructed from the

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<sup>2</sup>The concept of equivalent martingale measure will be generalised in Chapter 3.

absence of arbitrage<sup>3</sup>, and not from the beliefs of the agents.

Using equation (1.1) we can establish the following result.

**Proposition 1.1.** *The price of an option is the expectation under  $\mathbb{Q}$  of its discounted value:*

$$q = \mathbb{E}^{\mathbb{Q}} \left[ \frac{C_1}{1+r} \right] = \frac{\pi C_h + (1-\pi)C_b}{1+r} \quad (\text{resp. } P_1, P_h, P_b) \quad (1.4)$$

From (1.4), we observe that

$$C_1 - P_1 = \max(S_1 - K, 0) - \max(K - S_1, 0) = S_1 - K$$

Discounting and applying (1.4) to each term, we have:

$$C - P = \mathbb{E}^{\mathbb{Q}} \left[ \frac{S_1 - K}{1+r} \right] = \frac{\mathbb{E}^{\mathbb{Q}}[S_1]}{1+r} - \frac{K}{1+r} = S - \frac{K}{1+r},$$

where we have used the martingale property (1.3) in the last equality.

**Proposition 1.2.** *Let  $C$  and  $P$  be the prices at  $t = 0$  of a European call and put with the same strike  $K$  and the same maturity  $t = 1$ . We have:*

$$C - P = S - \frac{K}{1+r}. \quad (1.5)$$

This relation, known as put-call parity, is remarkable in that it involves no assumption about the price of the underlying at maturity  $S_1$ , no volatility model, and does not depend on  $\pi$ : it is a purely structural equation, valid in any no-arbitrage model in which the call and the put share the same underlying, the same strike, and the same maturity. Furthermore, it is all the more significant in that it connects the two types of options: knowing the price of the call immediately determines that of the put, and vice versa.

The smooth functioning of these relations rests entirely on the fundamental economic concept of the absence of arbitrage opportunities. It is necessary to rigorously formalise what an arbitrage is within the framework of our model.

**Definition 1.4.** *An arbitrage opportunity arises if a portfolio  $(\alpha, \beta)$  satisfies one of the following two conditions:*

1.  $\alpha + \beta S < 0$  and  $\alpha(1+r) + \beta S_1 \geq 0$
2.  $\alpha + \beta S = 0$  and  $\alpha(1+r) + \beta S_1 > 0$

In other words, an arbitrage opportunity arises if there exists a portfolio whose initial value is negative or zero and whose terminal value is strictly positive. It is a "risk-free gain".

The *no-arbitrage opportunity* (NAO) postulates that no arbitrage exists in the market. This assumption is economically natural: if an arbitrage opportunity were to exist, all agents would exploit it instantaneously, causing it to disappear through price adjustment.

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<sup>3</sup>This concept will be formalised below.



Moreover, the following characterisation can be established: the NAO assumption holds if and only if

$$S_b < (1 + r)S < S_h$$

We can now show that under the NAO assumption, the unique option price that does not give rise to an arbitrage opportunity is given by (1.1). This allows us to justify the option price obtained earlier.

*Proof.* Cf. *Financial Markets in Continuous Time* page 11 (French edition)/page 5 (English edition).  $\square$

In order to complete the description of this market environment, it is appropriate to introduce the notion of volatility of the underlying. This analysis requires defining risk measures associated with the asset, reasoning under a subjective probability  $p \in [0, 1]$  representing the likelihood of being in the up state (which, as a reminder, is distinct from the risk-neutral probability  $\pi$ ). The expectation under this measure is denoted  $\mathbb{E}^p$ .

By definition, the *rate of return* of the stock is given by  $R = (S_1 - S)/S$ . Its expectation under  $\mathbb{P}$  is

$$m_S = \mathbb{E}^{\mathbb{P}}[R] = \frac{pS_h + (1 - p)S_b}{S}$$

To measure the risk associated with the asset, we use the *volatility* of the stock. This is the standard deviation of its rate of return:

$$v_S = \sqrt{\text{Var}^{\mathbb{P}}(R)} = \frac{S_h - S_b}{S} \sqrt{p(1 - p)}$$

We will return to these notions in the following chapters.

This two-state modelling of nature (rise or fall) constitutes the ideal framework for introducing the basic concepts, but it raises a fundamental question: what happens if  $S_1$  can take  $k > 2$  values  $s_1 < s_2 < \dots < s_k$ ?

The replication system for a call  $(c_1, \dots, c_k)$  becomes:

$$\alpha(1 + r) + \beta s_j = c_j, \quad j = 1, \dots, k.$$

This is a system of  $k$  equations in 2 unknowns  $(\alpha, \beta)$ . For  $k > 2$ , it is over-determined and therefore has no solution: it is no longer possible to replicate the option. The market is then said to be *incomplete*. On the other hand, referring back to the construction of the risk-neutral probability (1.2), we observe that there no longer exists one but several risk-neutral probabilities.

We define the set of risk-neutral measures as follows:

$$\mathcal{P} := \left\{ \mathbb{Q} = (q_1, \dots, q_k) \mid \forall j : q_j > 0, \sum_{j=1}^k q_j = 1, \sum_{j=1}^k q_j s_j = (1 + r)S \right\}.$$

The last condition corresponding to the martingale property (1.3).

We denote by  $[S_b, S_h]$  the interval representing all possible prices at  $t = 1$ . Suppose now that  $S_1$  is a random variable taking its values in  $[S_b, S_h]$  and that  $S_b \leq (1 + r)S \leq S_h$ . The set  $\mathcal{P}$  of equivalent risk-neutral measures is infinite-dimensional (the replication system being over-determined). We then have the following result, which specifies the bounds of the range for an option with a convex payoff (such as  $C_1$  for example).

**Proposition 1.3.** *Let  $g : \mathbb{R} \rightarrow \mathbb{R}$  be a convex function (for example  $g(x) = (x - K)^+$ ). Then:*

$$\begin{aligned} \sup_{\mathbb{Q} \in \mathcal{P}} \mathbb{E}^{\mathbb{Q}} \left[ \frac{g(S_1)}{1+r} \right] &= \frac{1}{1+r} \left[ g(S_h) \frac{(1+r)S - S_b}{S_h - S_b} + g(S_b) \frac{S_h - (1+r)S}{S_h - S_b} \right] \\ &= \frac{1}{1+r} [g(S_h)\pi + g(S_b)(1-\pi)] \end{aligned}$$

*If  $g$  is of class  $C^1$ , we have:*

$$\inf_{\mathbb{Q} \in \mathcal{P}} \mathbb{E}^{\mathbb{Q}} \left[ \frac{g(S_1)}{1+r} \right] = \frac{g((1+r)S)}{1+r}.$$

*Proof.* Cf. *Financial Markets in Continuous Time* page 15 (French edition)/page 11 (English edition)  $\square$

A notable remark: we can interpret this result in terms of volatility. Indeed, the upper bound is attained when  $S_1$  equals  $S_b$  or  $S_h$ , that is, when the variance is maximal. On the other hand, the lower bound is attained when  $S_1$  equals  $(1+r)S$ , in other words when the variance is zero. We may then conclude that the interval  $[S_b, S_h]$  reflects the uncertainty about the volatility of the underlying asset. This is an important result, as it shows that even without a precise volatility model, one can still bound the price of an option.

This first chapter has established the conceptual foundations of arbitrage pricing. Two major lessons emerge and will structure the remainder of the dissertation. First, the price of a derivative is the expectation under the risk-neutral measure  $\mathbb{Q}$ , which is not a belief but a mathematical construction guaranteeing consistency with the NAO assumption. Second, in a complete market (here, two assets for  $k = 2$  states), the risk-neutral measure is unique and the price is unique. In contrast, in an incomplete market ( $k > 2$ ), the risk-neutral measure is no longer unique and there is no canonical price, but rather a range thereof.

# Chapter 2

## The general one-period model: $(d + 1)$ assets and $k$ states of the world

Chapter 1 established the foundations of arbitrage pricing theory in the simplest possible setting: two assets and two states of the world. We now turn to the case of a one-period market with  $(d + 1)$  assets and  $k$  states of the world. The aim of this chapter is not generalisation per se, but rather the illumination of the underlying mathematical architecture.

To study this expanded structure, we consider a market with two dates,  $t = 0$  and  $t = 1$ , on which  $d + 1$  assets indexed by  $i \in \{0, 1, \dots, d\}$  are traded. The world may find itself at  $t = 1$  in one of  $k$  states, indexed by  $j \in \{1, \dots, k\}$ . In order to handle these variables efficiently, we introduce the following matrix notation:

- $S = (S^0, S^1, \dots, S^d)^\top \in \mathbb{R}^{d+1}$  denotes the column vector of asset prices at the initial date  $t = 0$ .
- $V \in \mathcal{M}_{k \times (d+1)}(\mathbb{R})$  represents the payoff matrix, each of whose coefficients  $v_j^i$  corresponds to the value of asset  $i$  in state  $j$  at maturity  $t = 1$ .

We will now define, in matrix terms, what a portfolio and a risk-free asset are.

**Definition 2.1.** A portfolio is a vector  $\theta = (\theta^0, \theta^1, \dots, \theta^d)^\top \in \mathbb{R}^{d+1}$ , where  $\theta^i$  denotes the quantity held of asset  $i$ . Its initial value is the scalar product

$$V_0(\theta) := S \cdot \theta = \sum_{i=0}^d \theta^i S^i,$$

and its terminal value in state  $j$  is

$$(V\theta)_j = \sum_{i=0}^d \theta^i v_j^i.$$

In matrix notation, the vector of terminal payoffs of portfolio  $\theta$  is  $V\theta \in \mathbb{R}^k$ .

Just as in the case of 2 assets and 2 states of the world, the  $\theta^i$ ,  $i = (0, 1, \dots, d)$  may be positive for a long position or negative for a short sale.

**Definition 2.2.** Asset 0 is said to be risk-free with rate  $r > 0$  if  $v_j^0 = 1 + r$  for all  $j \in \{1, \dots, k\}$ , meaning its price is certain in every state of the world. Its initial value is normalised to  $S^0 = 1$ . The quantity  $\frac{1}{1+r}$  is the discount factor: it is the price to pay at  $t = 0$  in order to receive one euro at  $t = 1$ .

A remark on notation: we adopt the convention that vectors are column vectors. The product  $V\theta$  is therefore a column vector of  $\mathbb{R}^k$ , the transpose of  $V$  is  $V^\top \in \mathcal{M}_{(d+1) \times k}(\mathbb{R})$ , and the state-price relation will be written  $S = V^\top \beta$ , where  $\beta \in \mathbb{R}^k$ .

We will now redefine arbitrage using matrix notation.

**Definition 2.3.** *An arbitrage opportunity is said to exist if there exists a portfolio  $\theta \in \mathbb{R}^{d+1}$  satisfying one of the following two conditions:*

- (i) *its initial value  $\theta \cdot S = \sum_{i=0}^d \theta_i S^i$  is strictly negative and its terminal value is non-negative in every state of the world, i.e.:*

$$\sum_{i=0}^d \theta_i v_j^i \geq 0, \quad j \in \{1, \dots, k\};$$

- (ii) *its initial value  $\theta \cdot S$  is negative or zero and its terminal value is non-negative in every state of the world and strictly positive in at least one state, i.e.:*

$$\sum_{i=0}^d \theta_i v_j^i \geq 0, \quad j \in \{1, \dots, k\} \text{ and } \exists j_0, \sum_{i=0}^d \theta_i v_{j_0}^i > 0.$$

The no-arbitrage opportunity (NAO) postulates that no portfolio satisfying these conditions exists.

To establish the link between this absence of arbitrage and the existence of a risk-neutral probability measure, it is necessary to rely on a result from convex geometry: the Minkowski separation theorem.

**Theorem 2.1.** *Let  $C_1$  and  $C_2$  be two non-empty disjoint convex sets in  $\mathbb{R}^m$ , with  $C_1$  closed and  $C_2$  compact. Then there exists a non-zero vector  $a \in \mathbb{R}^m$  and two real numbers  $b_1 < b_2$  such that*

$$\forall x \in C_1, \forall y \in C_2 : \quad \sum_{j=1}^m a_j x_j \leq b_1 < b_2 \leq \sum_{j=1}^m a_j y_j$$

Its content may be summarised as follows: two disjoint compact convex sets can always be separated by a hyperplane. The proof of the fundamental theorem below will consist in identifying the two convex sets to be separated. This theorem establishes an equivalence between the absence of arbitrage and the existence of a system of state prices.

**Theorem 2.2.** *The NAO assumption is equivalent to the existence of a vector  $\beta = (\beta_1, \dots, \beta_k)^\top \in \mathbb{R}_{++}^k$  with strictly positive components, called the **state-price vector**, such that:*

$$S^i = \sum_{j=1}^k v_j^i \beta_j \quad \forall i \in \{0, \dots, d\} \quad (2.1)$$

*In particular:*

$$S = V^\top \beta$$

*Proof.* We prove the direct implication. Let  $S^\top$  be the row vector of initial prices  $(S^0, S^1, \dots, S^d)$  and define  $U$  as the vector subspace of  $\mathbb{R}^{k+1}$  formed by the discounted net cash flows of portfolios:

$$U := \left\{ z \in \mathbb{R}^{k+1} \mid z = \begin{pmatrix} -S^\top \\ V \end{pmatrix} x; \ x \in \mathbb{R}^{d+1} \right\}$$

where  $x$  represents the composition of a portfolio. The NAO assumption implies that the subspace  $U$  cannot intersect the set of non-zero positive payoffs, for otherwise this would mean that initial prices are negative or zero while payoffs are non-negative and strictly positive in at least one state. The only feasible portfolio is  $x = 0$ , and thus:

$$U \cap \mathbb{R}_+^{k+1} = \{0\}$$

In particular:

$$U \cap \Delta_k = \emptyset$$

where  $\Delta_k$  is the standard simplex of  $\mathbb{R}^{k+1}$ :

$$\Delta_k = \left\{ w \in \mathbb{R}_+^{k+1} \left| \sum_{j=0}^k w_j = 1 \right. \right\}.$$

By the Minkowski separation theorem, since  $U$  is a closed convex vector subspace (a finite-dimensional subspace being the image of a linear map) and  $\Delta_k$  is a compact convex set (closed and bounded by the column vector  $1 \in \mathbb{R}^{k+1}$ ) disjoint from  $U$ , there exists a vector of coefficients  $\{\beta_j; j = 0, \dots, k\}$ , not all zero, and two constants  $b_1$  and  $b_2$  such that:

$$\sum_{j=0}^k \beta_j z_j \leq b_1 < b_2 \leq \sum_{j=0}^k \beta_j v_j \quad \forall z \in U, \forall v \in \Delta_k$$

Since  $0 \in U$ , we have  $b_1 \geq 0$ . By choosing  $v$  to be the vector whose components are all zero except the  $j$ -th, which equals 1, we deduce that  $\beta_j > 0$  for all  $j \in \{0, \dots, k\}$ . Up to normalisation and without loss of generality, we may set  $\beta_0 = 1$ .

We then define the state-price vector by  $\beta = (\beta_1, \dots, \beta_k)^\top$ . Substituting back the form of the elements  $z \in U$  (where  $z_0 = -S \cdot x$  and  $z_j = (Vx)_j$ ), the inequality  $z_0 + \sum_{j=1}^k \beta_j z_j = 0$  rewrites in matrix form as:

$$(-S + V^T \beta) \cdot x \leq 0, \quad \forall x \in \mathbb{R}^{d+1}$$

Since this equality holds for every portfolio  $x$ , it also holds for  $-x$ , from which we deduce  $S = V^T \beta$ . Written component by component (for each asset  $i \in \{0, \dots, d\}$ ):

$$S^i = \sum_{j=1}^k \beta_j v_j^i \quad \text{with } \beta_j > 0, \forall j \in \{1, \dots, k\}$$

The proof of the converse is straightforward: if such strictly positive state prices exist, any portfolio with non-positive initial cost ( $S \cdot x \leq 0$ ) cannot generate strictly positive future payoffs in every state of the world without violating the equality  $S \cdot x = \beta^T Vx$ .  $\square$

Consider state  $j \in \{0, \dots, k\}$ . The scalar  $\beta_j$  is called the *state price* of state  $j$ : it is the price at  $t = 0$  of an asset that pays 1 euro at  $t = 1$  if state  $j$  occurs, and 0 in all other states. Such an asset is called an *Arrow-Debreu asset*. In other words, if there exists a portfolio  $\theta_j$  whose payoff is:

$$\delta_{\omega,j} = \begin{cases} 1 & \text{if } \omega = j \\ 0 & \text{otherwise} \end{cases}$$

then its price at  $t = 0$  will be  $\beta_j$ . In particular:  $V\theta^{(j)} = (\delta_{1,j}, \dots, \delta_{k,j})$ .

Concretely,  $\beta_j$  represents the market price of one euro delivered specifically in state of the world  $j$ . Furthermore, relation (2.1) shows that the price of any asset  $i$  is the sum of its future values  $v_j^i$  weighted by its state prices. Thus, for each state, one multiplies what the asset yields ( $v_j^i$ ) by the price of one euro in that state ( $\beta_j$ ).

In general, the vector  $\beta$  is not unique: the equation  $S = V^\top \beta$  is a system of  $d + 1$  equations in  $k$  unknowns. If  $k > d + 1$ , the system is under-determined and may admit infinitely many solutions. This will indeed be the case in an incomplete market. However, in a complete market the solution is unique, as we shall see later.

We will now introduce probabilities into the model. To do so, we will exploit the NAO assumption and Theorem (2.1). As we shall see, the existence of state prices (being strictly positive) allows us to construct a risk-neutral measure.

**Proposition 2.1.** *Under the NAO assumption, if asset 0 is risk-free with rate  $r > 0$ , there exists a probability measure  $\mathbb{Q}$  on the states of the world  $\{1, \dots, k\}$  with strictly positive weights, such that for every asset  $i \in \{1, \dots, d\}$ :*

$$S^i = \frac{1}{1+r} \sum_{j=1}^k \mathbb{Q}(j) v_j^i = \mathbb{E}^{\mathbb{Q}} \left[ \frac{S_1^i}{1+r} \right] \quad (2.2)$$

*Proof.* The risk-free asset satisfies  $v_j^0 = 1 + r$  for all  $j$ , and  $S^0 = 1$ . Relation (2.1) for  $i = 0$  gives:

$$1 = \sum_{j=1}^k \beta_j (1+r) \implies \sum_{j=1}^k \beta_j = \frac{1}{1+r}.$$

Set  $\pi_j := (1+r)\beta_j$ . Then  $\pi_j > 0$  for all  $j$  and  $\sum_{j=1}^k \pi_j = 1$ : the  $\pi_j$  define a probability measure  $\mathbb{Q}$  on  $\{1, \dots, k\}$ . For  $i \in \{1, \dots, d\}$ , using relation (2.1) again:

$$S^i = \sum_{j=1}^k v_j^i \beta_j = \sum_{j=1}^k v_j^i \frac{\pi_j}{1+r} = \frac{1}{1+r} \mathbb{E}^{\mathbb{Q}}[S_1^i]. \quad \square$$

It is once again important to note that for all  $j \in \{1, \dots, k\}$ ,  $\mathbb{Q}(j) = \pi_j$  does not represent the probability of being in state  $j$ . It is a measure arising from the NAO assumption, which serves to determine the price of our option.

It is possible to reformulate Theorem 2.2 in probabilistic terms. The measure  $\mathbb{Q}$  from the preceding Proposition may also be called the *equivalent martingale measure*. That is, under  $\mathbb{Q}$ , discounted prices are martingales and  $\mathbb{Q}$  is equivalent to  $\mathbb{P}$  (meaning  $\mathbb{Q}(j) > 0$  for all  $j$ ), where  $\mathbb{P}$  represents the probability of being in state of the world  $j$ . This is the discrete one-period version of the first fundamental theorem of asset pricing, whose generalisation to continuous time constitutes one of the pillars of modern mathematical finance.

We will now redefine, in matrix terms, what a replicable asset and a complete market are.

**Definition 2.4.** *Let  $z \in \mathbb{R}^k$  be an asset. Under the NAO assumption,  $z$  is said to be replicable if there exists a portfolio  $\theta \in \mathbb{R}^{d+1}$  such that  $V\theta = z$ , that is*

$$\sum_{i=0}^d \theta^i v_j^i = z_j \quad \forall j \in \{1, \dots, k\}.$$

**Definition 2.5.** *The market is said to be complete if every asset  $z \in \mathbb{R}^k$  is replicable, that is, if there exists a portfolio  $\theta$  such that  $V\theta = z$ .*

*A market is complete if one can achieve, with a portfolio chosen at  $t = 0$ , any wealth vector at  $t = 1$ .*

Translated into terms of linear algebra, this completeness property amounts to ensuring that the linear map associated with the payoff matrix is capable of spanning the entire space of states of nature. This leads us directly to the following algebraic characterisation:

**Proposition 2.2.** *The market is complete if and only if the matrix  $V$  has rank  $k$ .*

*Proof.* The matrix  $V$  has rank  $k$  if and only if  $\text{Im}(V) = \mathbb{R}^k$ , that is,  $V$  is surjective. Thus, the equation  $V\theta = z$  has at least one solution, so every asset is replicable.  $\square$

This result imposes a constraint on the market. It can be complete only if it has at least as many assets as possible states of the world ( $d + 1 \geq k$ ).

We will now show that in a no-arbitrage market, completeness guarantees not only the existence, but also the uniqueness of the risk-neutral measure.

**Proposition 2.3.** *In a complete no-arbitrage market, the state-price vector  $\beta$  and the risk-neutral measure  $\mathbb{Q}$  exist and are unique.*

*Proof.* In a complete market,  $\text{rg}(V) = k$ , so  $V^\top$  has rank  $k$ , which means the map  $\beta \mapsto V^\top \beta$  is injective. The equation  $V^\top \beta = S$  therefore admits at most one solution. Since the NAO assumption guarantees the existence of a solution with strictly positive components, this solution is unique. The uniqueness of  $\mathbb{Q}$  follows from the relation  $\mathbb{Q}(j) = \pi_j = (1 + r)\beta_j$ .  $\square$

This result is fundamental in option pricing. The uniqueness of the risk-neutral measure, ensured by the completeness of the market, allows us to perfectly replicate any asset, making it possible to assign a “fair” and unique price to any option. Moreover, existence, guaranteed by the NAO assumption, asserts that this price is “consistent” with the laws of the market. Finally, the existence and uniqueness of state prices ensure that every future cash flow can be precisely valued ( $\beta_j$ ) and perfectly hedged by a replicating portfolio.

In particular, in a complete market, we can construct a hedging portfolio that replicates the Arrow-Debreu asset. Thus, there exists a portfolio  $\theta^{(j)}$  such that  $V\theta^{(j)} = (\delta_{1,j}, \dots, \delta_{k,j}) = e_j$ , where  $e_j$  is the  $j$ -th vector of the canonical basis of  $\mathbb{R}^k$ . Its initial value is, by relation (2.1):

$$S \cdot \theta^{(j)} = (V^\top \beta) \cdot \theta^{(j)} = \beta^\top (V\theta^{(j)}) = \beta^\top e_j = \beta_j.$$

Thus,  $\beta_j$  is indeed the price of the Arrow-Debreu asset in state  $j$ . In fact, an Arrow-Debreu asset can be seen as a “pure risk”. It is the elementary unit of uncertainty. It corresponds to a scenario in which one earns 1 euro in one specific state  $j$  and 0 euros everywhere else. To purchase this risk is to bet on the realisation of a single scenario.

In a no-arbitrage market, if two portfolios  $\theta$  and  $\theta'$  replicate the same asset  $z$  (i.e.,  $V\theta = V\theta' = z$ ), then they have the same initial value:  $S \cdot \theta = S \cdot \theta'$ .

*Proof.* The portfolio  $\theta - \theta'$  satisfies  $V(\theta - \theta') = 0$ . By condition (i) of the NAO assumption:  $S \cdot (\theta - \theta') = 0$ , from which  $S \cdot \theta = S \cdot \theta'$ .  $\square$

This result is crucial as it guarantees that the *replication price*  $(S \cdot \theta)$  is well-defined, that is, it does not depend on the choice of replicating portfolio.

We will now state and prove the theorem allowing us to find the “fair” price of an asset at  $t = 0$  in the case of  $d + 1$  assets and  $k$  states of the world.

**Proposition 2.4.** *In a complete no-arbitrage market, the price at  $t = 0$  of any asset  $z \in \mathbb{R}^k$  is:*

$$\sum_{j=1}^k \beta_j z_j = \frac{1}{1+r} \sum_{j=1}^k \mathbb{Q}(j) z_j = \mathbb{E}^{\mathbb{Q}} \left[ \frac{z}{1+r} \right]. \quad (2.3)$$

*Proof.* Let  $\theta$  be a portfolio replicating  $z$ :  $V\theta = z$ . Using (2.1) and (2.2), its initial value is:

$$S \cdot \theta = (V^\top \beta) \cdot \theta = \beta^\top (V\theta) = \beta^\top z = \sum_{j=1}^k \beta_j z_j.$$

□

Returning to our main objective, namely the pricing of an option, we can conclude with the following result. In a complete no-arbitrage market, the price at  $t = 0$  of a European call with strike  $K$  on asset  $i$  is:

$$C = \frac{1}{1+r} \sum_{j=1}^k \mathbb{Q}(j) \max(v_j^i - K, 0) = \mathbb{E}^{\mathbb{Q}} \left[ \frac{(S_1^i - K)^+}{1+r} \right]. \quad (2.4)$$

This formula is the direct generalisation of (1.4) from Chapter 1.

This chapter has generalised the results of Chapter 1 to a complete algebraic architecture. The conceptual advances can be summarised in two points: first, the NAO assumption ensures the existence of the risk-neutral measure, and second, in a complete market, this measure is unique and every asset admits a well-defined initial price.

Chapter 3 will introduce the temporal dimension: instead of a single period, we will consider  $N$  successive periods, and the concepts of this chapter (NAO, martingale measure, replication price, completeness) will be applied recursively at each node of a tree. It is this recursion that will lead, in Chapter 6, to the Black-Scholes formula.



# Chapter 3

## Dynamic models in discrete time: general framework

The transition from Chapter 2 to Chapter 3 is a transition from linear algebra to discrete-time probability theory. In Chapter 2, a market was entirely described by two matrices,  $S$  and  $V$ , with no temporal structure. From now on, the market comprises  $N + 1$  dates and prices are stochastic processes whose evolution is governed by a growing flow of information, represented by a filtration.

Two new and fundamental concepts make their appearance: the *self-financing strategy*, which formalises the absence of fund injections or withdrawals, and the *martingale*, which is the natural probabilistic object for expressing the consistency of discounted prices over time. We will prove the fundamental theorem in finite horizon: the NAO assumption is equivalent to the existence of an equivalent martingale measure, and the market is complete if and only if this measure is unique. The tree representation will provide a bridge between this dynamic framework and the one-period results of the preceding chapters: the NAO condition in the global sense reduces to the NAO condition at each node of the tree. This architecture will lead to the Cox-Ross-Rubinstein binomial model, whose option pricing and closed-form formula we will study in detail.

We place ourselves on a probability space  $(\Omega, \mathcal{F}, P)$  equipped with a *filtration*  $(\mathcal{F}_n)_{n=0}^N$ , that is, an increasing family of sub- $\sigma$ -algebras of  $\mathcal{F}$ :

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_N = \mathcal{F}$$

We use this mathematical object because it has the advantage of being particularly realistic. Indeed, for every  $n$ , the  $\sigma$ -algebra  $\mathcal{F}_n$  represents the information available at date  $n$ : an event  $A \in \mathcal{F}_n$  is an event about which it is known, at date  $n$ , whether or not it has occurred. The growth of the filtration then reflects the fact that the past is not forgotten; there is no loss of information. The initial  $\sigma$ -algebra  $\mathcal{F}_0$  is the trivial  $\sigma$ -algebra  $\{\emptyset, \Omega\}$ : at date 0, nothing is known (beyond certain facts). We will now define what an adapted and predictable sequence of random variables is.

**Definition 3.1.** A sequence of random variables  $(S_n)_{n=0}^N$  is said to be  $(\mathcal{F}_n)$ -adapted if for every  $n \leq N$ , the random variable  $S_n$  is  $\mathcal{F}_n$ -measurable. This means that the information known at date  $n$  contains knowledge of the variable  $S_n$ .

**Definition 3.2.** A sequence of random variables  $(\theta_n)_{n=1}^N$  is said to be predictable (or  $(\mathcal{F}_{n-1})$ -adapted) if for every  $n \geq 1$ , the random variable  $\theta_n$  is  $\mathcal{F}_{n-1}$ -measurable.

This means that the decision made at  $n$  is based on the information available strictly before  $n$ . We will now “set the scene” by describing the market and exhibiting all the notation used.

The market consists of  $d+1$  assets whose prices at date  $n$  are given by a random vector  $S_n = (S_n^0, S_n^1, \dots, S_n^d)^\top \in \mathbb{R}_+^{d+1}$ , that is, each component is a random variable taking values in  $\mathbb{R}_+$ . The sequence  $(S_n)_{n=0}^N$  is assumed to be  $(\mathcal{F}_n)$ -adapted.

We also assume, as in Chapter 2, that asset 0 is risk-free with a constant rate of return  $r > 0$ , so that:

$$\begin{cases} S_0^0 = 1 & \text{for } n = 0 \\ S_n^0 = (1+r)^n & \forall n \geq 1 \end{cases}$$

Thus,  $S_n^0 = (1+r)^n$  is the discount factor. We denote the sequence of discounted prices as follows:

$$\forall n \in [0, N], \quad \hat{S}_n^i := \frac{S_n^i}{(1+r)^n}, \quad i \in \{0, \dots, d\}$$

By definition,  $\hat{S}_n^0 = 1$  for all  $n$ . We also adopt the convention  $\Delta S_n := S_n - S_{n-1}$  for any sequence of random variables  $(S_n)$ .

**Definition 3.3.** *A portfolio strategy is a predictable sequence of random vectors  $\theta = (\theta_n)_{n=1}^N$  taking values in  $\mathbb{R}^{d+1}$ , where  $\theta_n^i$  represents the number of units of asset  $i$  held between dates  $n-1$  and  $n$ . In other words, for every  $n \in [1, N]$ ,  $\theta_n = (\theta_n^0, \theta_n^1, \dots, \theta_n^d)$  is defined such that:*

$$\forall 1 \leq n \leq N, \quad \forall i \geq 0, \quad \theta_n^i \text{ is } \mathcal{F}_{n-1}\text{-measurable}$$

Moreover, the value of the portfolio at date  $n \geq 1$  is:

$$V_n(\theta) := \theta_n \cdot S_n = \sum_{i=0}^d \theta_n^i S_n^i$$

and its initial value is  $V_0(\theta) := \theta_1 \cdot S_0$ .

This definition once again reflects reality very faithfully. Indeed, predictability means that the investor chooses their portfolio on the basis of past information. This is the probabilistic translation of the fact that they cannot see into the future. We may note that, as in the preceding chapters, we impose no sign restriction on  $\theta_n^i$ , which allows for short sales or borrowing.

**Definition 3.4.** *A portfolio strategy  $\theta$  is self-financing if:*

$$\forall n \in \{1, \dots, N-1\}, \quad \theta_n \cdot S_n = \theta_{n+1} \cdot S_n$$

This means that the value of the portfolio after date  $n$  and strictly before date  $n+1$  (namely,  $\theta_n \cdot S_n$ ) equals the value of the next portfolio that the investor will constitute before prices change (namely,  $\theta_{n+1} \cdot S_n$ ), that is, before  $S_n$  moves to  $S_{n+1}$ . In other words, the portfolio rebalancing between  $n$  and  $n+1$  is financed entirely by the assets held, without any external injection or withdrawal of funds, and transactions are carried out without costs. One constructs a new portfolio using only the portfolio one already holds.

The self-financing condition can be reformulated very conveniently in terms of the variation in portfolio value. Indeed, the strategy  $\theta$  is self-financing if and only if, for every  $n \geq 1$ :

$$V_n(\theta) = V_{n-1}(\theta) + \theta_n \cdot \Delta S_n \quad (3.1)$$

In particular, the variations in portfolio value between two consecutive dates arise solely from the price variations of the assets held.

*Proof.* For  $n \geq 2$ , the self-financing condition at  $n - 1$  gives  $\theta_{n-1} \cdot S_{n-1} = \theta_n \cdot S_{n-1}$ . Therefore:

$$V_n(\theta) - V_{n-1}(\theta) = \theta_n \cdot S_n - \theta_{n-1} \cdot S_{n-1} = \theta_n \cdot S_n - \theta_n \cdot S_{n-1} = \theta_n \cdot \Delta S_n$$

□

By induction on (3.1), the value of the portfolio at any instant  $n$  is therefore:

$$V_n(\theta) = V_0(\theta) + \sum_{k=1}^n \theta_k \cdot \Delta S_k \quad (3.3)$$

It is often more convenient to work with the discounted prices  $\hat{S}_n^i = S_n^i / S_n^0$ . We denote by  $\hat{S}$  the vector of discounted prices. We define the *discounted value* of the portfolio by  $\hat{V}_n(\theta) := V_n(\theta) / S_n^0$ . For a self-financing strategy  $\theta$ , by virtue of relation (3.1), we see that the discounted value satisfies, for  $n > 0$ :

$$\hat{V}_n(\theta) = V_0(\theta) + \sum_{k=1}^n \theta_k \cdot \Delta \hat{S}_k$$

and  $\hat{V}_0(\theta) = V_0(\theta)$ . In particular:

$$\hat{V}_{n+1}(\theta) = \hat{V}_n(\theta) + \theta_{n+1} \cdot \Delta \hat{S}_{n+1} \quad (3.2)$$

It is important to note that the set of arbitrage opportunities is identical for the random vectors  $S$  and  $\hat{S}$ . Indeed, a strategy with zero initial value and strictly positive terminal value in nominal terms is also so in discounted terms (since  $S_n^0 > 0$  for all  $n$ ). We may therefore work indifferently with nominal or discounted prices depending on the context.

We will now define what an arbitrage opportunity is in a multi-date model with finite horizon.

**Definition 3.5.** *A self-financing strategy  $\theta$  is an arbitrage opportunity if:*

$$(i) \ P(V_0(\theta) = 0) = 1$$

$$(ii) \ P(V_N(\theta) \geq 0) = 1 \quad \text{and} \quad P(V_N(\theta) > 0) > 0$$

where  $P$  is the reference probability of the triple  $(\Omega, \mathcal{F}, P)$ . In other words, the investor starts with zero capital and ends at  $N$  with a profit that is non-negative and strictly positive on a set of strictly positive measure.

Once again, the NAO assumption postulates that no strategy satisfying this definition exists. We may note that two equivalent measures  $P$  and  $Q$  (in the sense that  $P(A) = 0 \Leftrightarrow Q(A) = 0$  for all  $A \in \mathcal{F}$ ) define exactly the same set of arbitrages. Thus, the NAO condition is an intrinsic property of the market, independent of the choice of reference probability among equivalent probabilities.

As already discussed in Chapter 2, we can show that there exists a martingale measure (which will be our future risk-neutral measure) if and only if the market is arbitrage-free.

**Theorem 3.1.** *The NAO assumption holds if and only if there exists a probability measure  $\mathbb{Q}$  equivalent to  $P$  under which the sequence of random vectors of discounted prices  $(\hat{S}_n^i)_{n=0}^N$  are  $(\mathcal{F}_n)$ -martingales for all  $i \in \{0, \dots, d\}$ . We denote by  $\mathcal{P}$  the set of these equivalent martingale measures.*

We will not prove this result, for the sake of expository clarity. The reader may refer to pages 44–45 of the French edition (or page 47 of the English edition) of *Financial Markets in Continuous Time*.

This result is absolutely fundamental. It transforms an economic and intuitive property (the impossibility of making money without risk and without an initial outlay) into a precise and tractable mathematical object (a martingale measure  $\mathbb{Q}$ ). It is by virtue of this theorem that the use of expectation calculations to determine the “fair price” of assets can be justified, since, as we shall see later, under this measure  $\mathbb{Q}$  (which is in fact the risk-neutral measure of the preceding chapters) the current price is simply the expectation of the discounted future cash flows. However, this does not guarantee the uniqueness of the price if we are not in a complete market.

We now assume that  $\Omega$  is finite. In this case, the filtration  $(\mathcal{F}_n)$  can be associated with a tree, which reduces the global study of the market to a succession of local problems, each identical to the one-period problem of the preceding chapters.

The  $\sigma$ -algebra  $\mathcal{F}_n$  is generated by a partition of  $\Omega$  into *atoms* (the minimal non-empty elements of  $\mathcal{F}_n$ ). An atom  $A_n^j \in \mathcal{F}_n$  represents the set of paths  $\omega \in \Omega$  that are indistinguishable at date  $n$ : to reach  $A_n^j$ , all paths have taken the same route. Each atom of  $\mathcal{F}_{n+1}$  is contained in exactly one atom of  $\mathcal{F}_n$ . In other words, for  $n = 1$ , we have:  $A_1^j = \bigcup_{i \in I_1^j} A_2^i$ , where  $I_1^j$  is the number of successors of node  $j$ .

We denote by  $\Delta^n(j)$  the set of nodes at date  $n + 1$  descending from  $j$ :

$$\Delta^n(j) := \left\{ A_{n+1}^\ell \in \mathcal{F}_{n+1} \mid A_{n+1}^\ell \subseteq A_n^j \right\}$$

Since the prices  $S_n$  are  $\mathcal{F}_n$ -adapted, they are well-defined and constant on each atom  $A_n^j$ . One may therefore speak of the price at node  $j$  at date  $n$ , denoted  $S_n(j)$ .

It is possible to define arbitrage on a tree as follows.

**Definition 3.6.** *We say that there is arbitrage for the tree at node  $j$  if there exist  $n$ ,  $0 \leq n < N$ ,  $A_n^j \in \mathcal{F}_n$  and  $\theta_{n+1} \in \mathbb{R}^{d+1}$  such that:*

$$\theta_{n+1} \cdot (\hat{S}_{n+1} - \hat{S}_n) \geq 0, \quad \forall \omega \in A_n^j$$

*with strict inequality for at least one successor  $A_{n+1}^\ell \in \Delta^n(j)$ .*

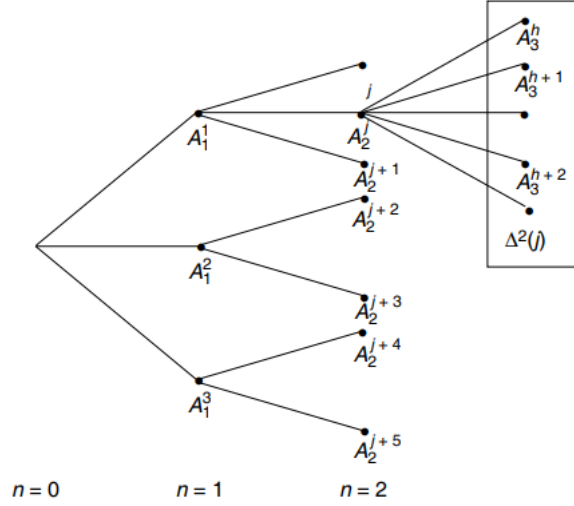


Figure 3.1: Representation of a general tree

**Proposition 3.1.** *The NAO condition holds for the model if and only if the NAO condition holds on the tree.*

*Proof.* Cf. *Financial Markets in Continuous Time*, page 51 (French edition)/page 53 (English edition).  $\square$

Proposition (3.1) is of considerable practical importance. It allows the verification of the global NAO condition to be reduced to a node-by-node verification, each local verification being identical to the one-period problem already solved in Chapter 2. It is this reduction that enables the explicit construction of the risk-neutral measure in the binomial model.

To arrive at the pricing calculations, it remains to define replicability and complete markets.

**Definition 3.7.** *An  $\mathcal{F}_N$ -measurable random variable  $X$  is replicable if there exists a self-financing strategy  $\theta$  such that  $V_N(\theta) = X$ . The strategy  $\theta$  is then called a hedging strategy for  $X$ .*

**Definition 3.8.** *A market is complete if every  $\mathcal{F}_N$ -measurable random variable  $X$  is replicable.*

In other words, using the definition of  $V_N(\theta)$ , a market is complete if every  $\mathcal{F}_N$ -measurable random variable  $X$  satisfies  $X = \sum_{i=0}^d \theta_N^i S_N^i$ , where  $\theta$  is a self-financing strategy.

This allows us to arrive at the second fundamental theorem: the uniqueness of the equivalent martingale measure in a complete market.

**Theorem 3.2.** *An arbitrage-free market is complete if and only if there exists a unique equivalent martingale measure  $\mathbb{Q}$  equivalent to  $P$  under which discounted prices are martingales.*

*Proof.* Cf. *Financial Markets in Continuous Time*, page 52 (French edition)/page 54 (English edition).  $\square$

This result is crucial as it stipulates that when we are in a complete market, since  $\mathbb{Q}$  is unique, the pricing formula (which we will see in the next section) yields a unique price for every asset. If  $\mathbb{Q}$  were not unique, there would be several possible prices for the same product.

**Proposition 3.2.** *Let  $X$  be a replicable random variable. The initial value  $V_0(\theta)$  of any hedging strategy  $\theta$  is independent of the choice of  $\theta$  and equal to:*

$$V_0(\theta) = \mathbb{E}^{\mathbb{Q}} \left[ \frac{X}{S_N^0} \right] \quad \text{for all } \mathbb{Q} \in \mathcal{P} \quad (3.3)$$

*Proof.* Let  $\theta^1$  and  $\theta^2$  be two self-financing strategies such that  $V_N(\theta^1) = V_N(\theta^2) = X$ , and suppose  $V_0(\theta^1) > V_0(\theta^2)$ . The strategy  $\theta^2 - \theta^1$  is self-financing, with zero terminal value and initial value  $V_0(\theta^2) - V_0(\theta^1) < 0$ . This is an arbitrage opportunity, contradicting the NAO assumption. Therefore  $V_0(\theta^1) = V_0(\theta^2)$ . The equality  $V_0(\theta) = \mathbb{E}^{\mathbb{Q}} \left[ \frac{X}{S_N^0} \right]$  follows from the fact that  $\hat{V}(\theta)$  is a  $\mathbb{Q}$ -martingale (Theorem (3.1)) and from the following relation:

$$V_0(\theta) = \hat{V}_0(\theta) = \mathbb{E}^{\mathbb{Q}}[\hat{V}_N(\theta)] = \mathbb{E}^{\mathbb{Q}} \left[ \frac{X}{S_N^0} \right]$$

□

This means that even if there were several different ways (several strategies  $\theta$ ) of hedging exactly the same future cash flow  $X$ , they would all have exactly the same initial cost  $V_0(\theta)$ . Moreover, this calculation must necessarily be carried out under a special probability measure  $\mathbb{Q}$  (the martingale measure), and not under the historical probability  $P$ . Under this measure, the average return of risky assets equals the risk-free rate. Furthermore, the formula applies even when the market is incomplete, since the result is the same for every  $\mathbb{Q} \in \mathcal{P}$ . This means that if an asset is replicable, regardless of which martingale measure is used to perform the calculation, one will always arrive at the same initial value  $V_0$ . In a complete market, we define the price at time  $n$  of the sequence of random variables  $(X_n)_{n=1}^N$  adapted to the filtration by:

$$V_n(\theta) = S_n^0 \mathbb{E}^{\mathbb{Q}} \left[ \sum_{t=1}^N \frac{X_t}{S_t^0} \middle| \mathcal{F}_n \right] \quad (3.4)$$

This is the dynamic generalisation of the formula from Chapter 2. In particular, the price of the European call with strike  $K$  and maturity  $N$  on asset  $S^1$  equals, at each date  $n$ :

$$C_n = (1+r)^n \mathbb{E}^{\mathbb{Q}} \left[ \frac{(S_N^1 - K)^+}{(1+r)^N} \middle| \mathcal{F}_n \right] \quad (3.5)$$

# Chapter 4

## Dynamic option pricing

This chapter descends from the abstract framework of Chapter 3 towards a concrete and computable model: the Cox-Ross-Rubinstein (CRR) binomial model, created by John C. Cox, Stephen A. Ross and Mark Rubinstein and published in 1979 in an article entitled “*Option Pricing: A Simplified Approach*”. This model is the simplest realisation of a recombining tree, which allows us to readily illustrate all the fundamental concepts presented previously and to derive a closed-form pricing formula.

In this chapter we will establish the pricing formula for a European call in the  $N$ -period binomial model, making explicit two computation methods: the *backward induction on the tree* and the *closed-form formula*. We will then study *American options* and introduce the notion of the *Snell envelope*. Furthermore, where relevant, we will exhibit the implementation of the concept being explained.

The  $N$ -period binomial model is defined by:

- a risk-free asset with rate  $r > 0$ , which depends neither on the date nor on the state of the world,
- a risky asset with initial price  $S > 0$ , whose price at each period may rise by a factor  $h > 0$  or fall by a factor  $b > 0$ , with  $b < 1 + r < h$ .

At date  $n$ , the price of the risky asset is given by:

$$S_n^{(j)} = S \cdot h^j \cdot b^{n-j} \quad 0 \leq j \leq n \quad (4.1)$$

where  $j \in \{0, 1, \dots, n\}$  is the number of up-moves occurring between 0 and  $n$ . For example, if the stock price has risen twice, then its price at  $n = 2$  will be  $Sh^2$ .

The space  $\Omega$  is the set of sequences of length  $N$  of symbols  $\{h, b\}$ . Each  $\sigma$ -algebra  $\mathcal{F}_n$  is generated by the first  $n$  symbols: the information at date  $n$  is exactly the trajectory  $(S_0, S_1, \dots, S_n)$  (we know what happened over the previous  $n - 1$  days).

It is important to note that the tree structure is *recombining*:  $h^j b^{n-j} = h^{j'} b^{n-j'}$  if and only if  $j = j'$ . For example, if the stock price first rises and then falls, it has the same value as if it had first fallen and then risen (since  $Shb = Sbh$ ).

We note that for every  $n$  and every  $j$ ,  $\Delta^n(j)$  contains only two elements (cf. Figure 4.1). Moreover, there are  $n + 1$  distinct nodes at date  $n$  (and not  $2^n$  as in a non-recombining tree), which reduces the computational complexity from  $O(2^N)$  to  $O(N^2)$ . Indeed, to compute the price at maturity  $N$ , we must traverse all possible paths. Thus, if the tree

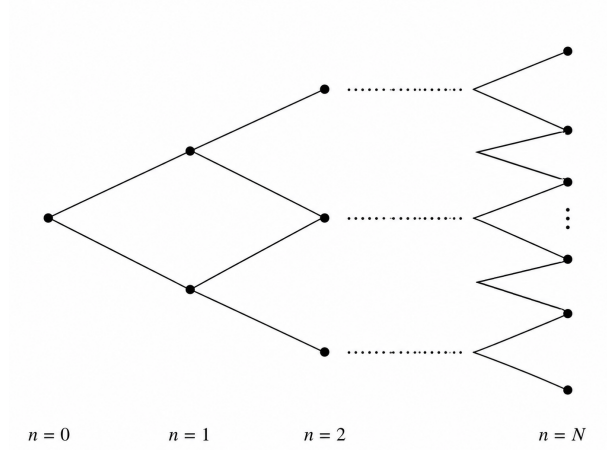


Figure 4.1: Recombining binomial tree

were not recombining, the number of computations required would be:

$$\sum_{n=0}^N 2^n = 2^{N+1} - 1$$

Whereas for the recombining tree we have:

$$\sum_{n=0}^N (n+1) = \frac{(N+1)(N+2)}{2}$$

The shift from exponential complexity in  $O(2^N)$  to quadratic complexity in  $O(N^2)$  is crucial. It transforms a problem whose computational cost becomes physically infeasible beyond 50 periods into an algorithm that can be executed for much larger values of  $N$ .

In matrix terms, we can encode this recombining tree structure in a lower triangular matrix  $M \in \mathcal{M}_{(N+1) \times (N+1)}(\mathbb{R})$  with:

$$M_{j,n} = S \cdot h^j \cdot b^{n-j}, \quad 0 \leq j \leq n \leq N$$

and  $M_{j,n} = 0$  for  $j > n$ . In the Python code, this corresponds to the function `price_tree`:

```
1 def price_tree(S, h, b, N):
2     M = np.zeros((N+1, N+1))
3     for i in range(N+1):           #columns = dates
4         for j in range(i+1):       #rows = number of up-moves
5             M[j, i] = S * (h**j) * (b**(i-j))
6     return M
```

By Proposition 3.1, the NAO condition is equivalent to the absence of arbitrage at each node. Applying the one-period result of Chapter 1 at each node, we obtain the *risk-neutral probability*:

$$\pi = \frac{(1+r) - b}{h - b} \quad (4.2)$$

This probability is well-defined and belongs to  $[0, 1]$  by virtue of the condition  $b < 1 + r < h$ . Moreover, since the parameters  $h$  and  $b$  are constants, it follows that the probability  $\pi$  is independent of the node and of the date.



The binomial market is complete since we have two assets and two successors at each node. Furthermore, this implies that every  $\mathcal{F}_N$ -measurable variable is replicable.

Since the binomial market is also arbitrage-free, by Theorem (3.2) there exists a unique equivalent martingale measure  $\mathbb{Q}$  on  $\Omega$  under which the transitions are i.i.d. Bernoulli with parameter  $\pi$ , that is:

$$\mathbb{Q}(\text{up}) = \pi, \quad \mathbb{Q}(\text{down}) = 1 - \pi$$

It then follows that for every  $n \in \{0, \dots, N\}$ ,  $S_n$  is a random variable following a binomial distribution with parameters  $n$  and  $\pi$ :

$$\mathbb{Q}(S_n = h^j b^{n-j} S) = \binom{n}{j} \pi^j (1 - \pi)^{n-j}$$

where:

$$\binom{n}{j} = \frac{n!}{j!(n-j)!}$$

We will now study the two methods that can be used to price a European option in the binomial model, beginning with the recursive backward induction on the tree.

**Method 1: backward recursive induction on the tree:** The general formula (3.4) takes the following form here, at date  $n$  and node  $j$ :

$$V_n^{(j)} = \frac{1}{1+r} \left[ \pi V_{n+1}^{(j+1)} + (1-\pi) V_{n+1}^{(j)} \right] \quad (4.3)$$

where  $V_{n+1}^{(j+1)}$  is the value at the node reached by an up-move (node  $(j+1, n+1)$ ) and  $V_{n+1}^{(j)}$  is that reached by a down-move (node  $(j, n+1)$ ).

*Proof.* At node  $(j, n)$ , the  $\sigma$ -algebra  $\mathcal{F}_n$  is represented by the atom  $\{J_n = j\}$ , where  $J_n$  denotes the number of up-moves that have occurred up to date  $n$ . The value of the portfolio at this particular node, denoted  $V_n^{(j)}$ , is written:

$$V_n^{(j)} = (1+r)^n \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_N)}{(1+r)^N} \mid J_n = j \right]$$

where  $\varphi$  is the payoff of the option ( $\varphi(x) = (x - K)^+$  for a call,  $(K - x)^+$  for a put).

Likewise, at date  $n+1$ , two states are accessible from node  $j$ . The respective portfolio values are written:

$$V_{n+1}^{(j+1)} = (1+r)^{n+1} \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_N)}{(1+r)^N} \mid J_{n+1} = j+1 \right]$$

for an up-move and:

$$V_{n+1}^{(j)} = (1+r)^{n+1} \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_N)}{(1+r)^N} \mid J_{n+1} = j \right]$$

for a down-move. Applying the tower property<sup>1</sup> by conditioning with respect to the  $\sigma$ -algebra  $\mathcal{F}_{n+1}$ , we can decompose the expectation of  $V_n^{(j)}$  as follows:

$$\mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_N)}{(1+r)^N} \mid J_n = j \right] = \mathbb{E}^{\mathbb{Q}} \left[ \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_N)}{(1+r)^N} \mid \mathcal{F}_{n+1} \right] \mid J_n = j \right]$$

---

<sup>1</sup>The tower property states that if  $\mathcal{G} \subset \mathcal{H}$ , then  $\mathbb{E}[\mathbb{E}[X \mid \mathcal{H}] \mid \mathcal{G}] = \mathbb{E}[X \mid \mathcal{G}]$

Given  $J_n = j$ , at the next step the price rises with probability  $\pi$  (giving  $J_{n+1} = j + 1$ ) or falls with probability  $1 - \pi$  (giving  $J_{n+1} = j$ ). Therefore:

$$\mathbb{E}^{\mathbb{Q}} \left[ \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_N)}{(1+r)^N} \middle| \mathcal{F}_{n+1} \right] \middle| J_n = j \right] = \pi \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_N)}{(1+r)^N} \middle| J_{n+1} = j + 1 \right] + (1-\pi) \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_N)}{(1+r)^N} \middle| J_{n+1} = j \right]$$

Using the definitions introduced previously, we isolate the conditional expectations at date  $n + 1$ :

$$\mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_N)}{(1+r)^N} \middle| J_{n+1} = j + 1 \right] = \frac{V_{n+1}^{(j+1)}}{(1+r)^{n+1}} \quad \text{and} \quad \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_N)}{(1+r)^N} \middle| J_{n+1} = j \right] = \frac{V_{n+1}^{(j)}}{(1+r)^{n+1}}$$

Substituting these expressions into the equation for  $V_n^{(j)}$ , we obtain:

$$V_n^{(j)} = (1+r)^n \left[ \pi \frac{V_{n+1}^{(j+1)}}{(1+r)^{n+1}} + (1-\pi) \frac{V_{n+1}^{(j)}}{(1+r)^{n+1}} \right]$$

Thus, simplifying:

$$V_n^{(j)} = \frac{1}{1+r} \left[ \pi V_{n+1}^{(j+1)} + (1-\pi) V_{n+1}^{(j)} \right]$$

□

It is important to highlight the distinction between the time index  $n$  and the state index  $j$  in this recurrence. Whereas the classical binomial price tree is often represented in a symmetric arborescent form, the formalisation by pairs  $(n, j)$  induces an asymmetric rectangular grid illustrated by Figure (4.2).

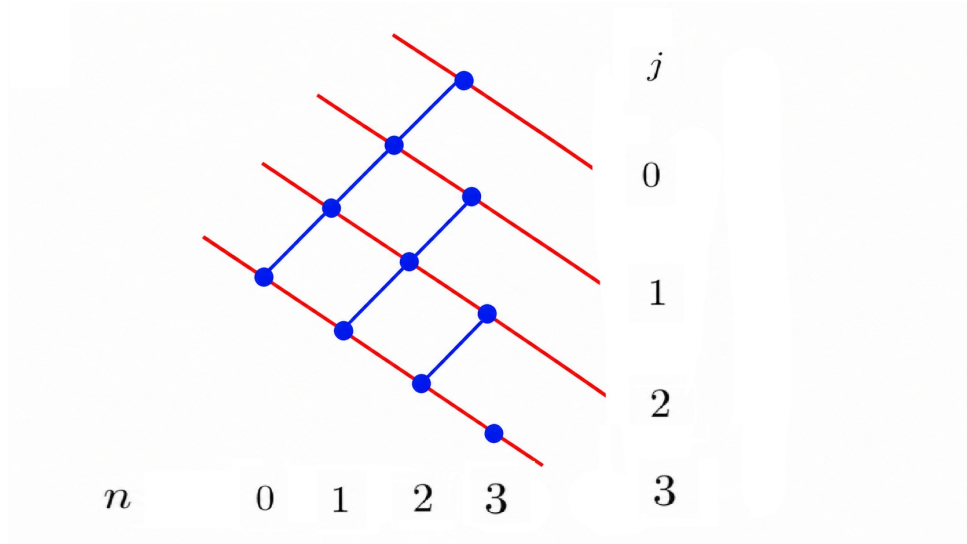


Figure 4.2: Graph illustrating the argument

Since the binomial tree is recombining, the evolution of the portfolio from date  $n$  to date  $n + 1$  follows the dynamics below:

- In the event of an up-move (probability  $\pi$ ): the underlying asset increases. By definition, the counter of the cumulative number of up-moves is incremented. A double change of indices then occurs:  $(n, j) \rightarrow (n + 1, j + 1)$ . Graphically, this corresponds to a diagonal displacement (change of level  $j$ ).

- In the event of a down-move (probability  $1 - \pi$ ): the underlying asset decreases. Time advances, but no up-move event is recorded. The counter  $j$  therefore remains unchanged. The transition is written:  $(n, j) \rightarrow (n + 1, j)$ . Graphically, the point moves horizontally along the same level line  $j$ .

This geometric asymmetry explains the structure of equation (4.3). The term  $V_{n+1}^{(j+1)}$  reflects the impact of an up-move on the portfolio (jump to the higher level), while the term  $V_{n+1}^{(j)}$  reflects the impact of a down-move (remaining on the same up-move trajectory).

The recurrence proceeds from maturity  $n = N$  back to  $n = 0$  as follows:

**Initialisation:** Recall that  $\varphi$  is the payoff of the option ( $\varphi(x) = (x - K)^+$  for a call,  $(K - x)^+$  for a put). For every  $j \in \{0, \dots, N\}$ :

$$V_N^{(j)} = \varphi(M_{j,N}) = \varphi(Su^j d^{N-j}), \quad (4.4)$$

The initialisation is nothing other than the payoff at maturity.

**Recurrence:** For  $n = N - 1, N - 2, \dots, 0$  and  $j = 0, 1, \dots, n$ :

$$V_n^{(j)} = \frac{\pi V_{n+1}^{(j+1)} + (1 - \pi) V_{n+1}^{(j)}}{1 + r}.$$

Thus, the price of the European call is  $C = V_0^{(0)}$ . This method is implemented in the function `option_value2`:

```

1 def option_value2(M, K, r, dt, pi, N, option_type, american=False):
2     V = np.zeros(np.shape(M))
3     for j in range(0, N+1):
4         V[j, N] = payoff(M[j, N], K, option_type)    # initialisation at
maturity
5     for i in range(N-1, -1, -1):                        # backward induction
on the tree
6         for j in range(0, i+1):
7             V[j, i] = np.exp(-r*dt) * ((pi * V[j+1, i+1]) + (1-pi) * V[j,
i+1])
8     return V[0, 0], V

```

It is crucial to note that, in this code, the discount factor is  $e^{-r \cdot dt}$  rather than  $(1+r)^{-1}$ , with  $dt = T/N$ . This corresponds to the CRR parametrisation (which we will see later)<sup>2</sup> where the rate  $r$  is the risk-free rate:  $e^{r \cdot dt} \approx 1 + r \cdot dt$  for small  $dt$ .

**Method 2: closed-form formula for European options:** For a European call, the recombining property of the tree makes it possible to avoid the recurrence and to express the price directly as a finite sum.

**Proposition 4.1.** *The price at  $t = 0$  of the European call with strike  $K$  and maturity  $N$  in the binomial model is:*

$$C = \frac{1}{(1+r)^N} \sum_{j=0}^N \binom{N}{j} \pi^j (1-\pi)^{N-j} (Sh^j b^{N-j} - K)^+ \quad (4.5)$$

---

<sup>2</sup>Cf. Chapter 5

*Proof.* By formula (3.3),  $C = \mathbb{E}^{\mathbb{Q}}[(S_N - K)^+ / B_N]$ . Thus, substituting the binomial distribution of  $S_N$  under  $\mathbb{Q}$ :

$$C = \frac{1}{(1+r)^N} \sum_{j=0}^N \binom{N}{j} \pi^j (1-\pi)^{N-j} (Su^j d^{N-j} - K)^+$$

□

This formula is implemented in `option_value1`:

```
1 def option_value1(S, K, r, T, h, b, pi, N, option_type):
2     total = 0
3     for j in range(0, N+1):
4         S_node = (h**j) * (b**(N-j)) * S
5         total += math.comb(N, j) * (pi**j) * ((1-pi)**(N-j)) * payoff(
6             S_node, K, option_type)
7     return np.exp(-r*T) * total
```

Of course, both methods yield the same price. The numerical verification is performed in the notebook, which displays both values and confirms their equality. Moreover, we can also extend put-call parity to the binomial model in finite horizon.

**Proposition 4.2.** *Let  $C$  and  $P$  be the prices at  $t = 0$  of a European call and put with the same strike  $K$  and maturity  $N$  in the binomial model. We have:*

$$C - P = S - \frac{K}{(1+r)^N}.$$

*Proof.* We have  $(S_N - K)^+ - (K - S_N)^+ = S_N - K$  for every  $S_N$ . By linearity of the expectation under  $\mathbb{Q}$  and the martingale property:

$$C - P = \mathbb{E}^{\mathbb{Q}} \left[ \frac{S_N - K}{(1+r)^N} \right] = \frac{\mathbb{E}^{\mathbb{Q}}[S_N]}{(1+r)^N} - \frac{K}{(1+r)^N} = S - \frac{K}{(1+r)^N}.$$

□

The function `test_parity` in the notebook verifies this identity numerically.

Unlike a European option, which can only be exercised at maturity  $N$ , an American option gives the holder the freedom to exercise their right at any instant  $\tau \in \{0, 1, \dots, N\}$ . Exercising at date  $\tau$  delivers an immediate payoff  $\varphi(S_\tau)$ . The holder's problem consists in maximising their gain by determining an optimal exercise strategy, or in other words, in knowing at which moment to exercise their right in order to maximise their profits.

We begin by defining what a *stopping time* is.

**Definition 4.1.** *A stopping time is a random variable  $\tau : \Omega \rightarrow \{0, 1, \dots, N\}$  such that  $\{\tau = n\} \in \mathcal{F}_n$  for every  $n$ . A stopping time represents an exercise strategy whose execution at each date  $n$  requires only the information available up to  $n$ .*

For example, if we hold an American option, we observe the binomial tree day by day. At date  $n$ , we know the current value of the stock and can then decide to exercise. This is an exercise strategy measurable with respect to the present: it is therefore a stopping time. By contrast, a perfect strategy consisting in exercising at the exact moment when

the stock reaches its absolute maximum over the entire duration of the contract is not a stopping time, since knowing whether the current price is the absolute maximum would require knowledge of future prices.

The price of an American option at time  $t = 0$  is defined by the following optimisation problem:

$$V_0^{\text{am}} = \sup_{\tau \text{ s.t.}} \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_\tau)}{(1+r)^\tau} \right] \quad (4.6)$$

We can explain this by drawing a parallel with European options. Indeed, the price of a European option is given by formula (3.3), which directly yields the expectation. Still for European options, since no choice is possible, the contract requires waiting until maturity  $N$ . Thus, one simply computes the average of what the option will yield at that date, then discounts it over the  $N$  periods. Here, since the holder has the right to choose the most convenient moment, the price at  $t = 0$  must consider all possible exercise scenarios and select the best one. This allows us to deduce the existence of the sup over stopping times. Of course, this assumes that agents are rational, meaning they wish to lose as little (and gain as much) money as possible.

For every  $n \in \{0, \dots, N-1\}$  and every  $j \in \{0, \dots, n\}$ , the optimal value  $V_n^{(j),\text{am}}$  at date  $n$  and node  $j$  satisfies the following backward recurrence in  $n$ :

$$V_n^{(j),\text{am}} = \max \left( \varphi(S_n^{(j)}), \frac{\pi V_{n+1}^{(j+1),\text{am}} + (1-\pi) V_{n+1}^{(j),\text{am}}}{1+r} \right) \quad (4.7)$$

The economic interpretation follows directly from the property of American options: at each node, the holder chooses between exercising immediately (payoff  $\varphi(S_n^{(j)})$ ) and continuing to hold the option (continuation value given by the conditional expectation  $\mathbb{E}^{\mathbb{Q}}[V_{n+1}/(1+r) \mid \mathcal{F}_n]$ ).

**Definition 4.2.** *The sequence of random variables  $(V_n^{\text{am}})_{n=0}^N$  defined by (4.7) with  $V_N^{\text{am}} = \varphi(S_N)$  is called the Snell envelope of the gain sequence  $(\varphi(S_n))_{n=0}^N$ .*

We will now establish three properties arising from this definition.

**Proposition 4.3.** 1.  $(V_n^{\text{am}})_{n=0}^N$  is a discounted  $\mathbb{Q}$ -supermartingale, that is:

$$\mathbb{E}^{\mathbb{Q}} \left[ \frac{V_{n+1}^{\text{am}}}{(1+r)} \mid \mathcal{F}_n \right] \leq V_n^{\text{am}}$$

2. The optimal stopping time  $\tau^*$  is the first instant at which the value of the option coincides with its immediate payoff:

$$\tau^* = \min\{n : V_n^{\text{am}} = \varphi(S_n)\}$$

3. For an American call without dividends, it is never optimal to exercise the option before maturity. The American value then coincides with the European value for every  $n$ :

$$V_n^{\text{am}} = V_n^{\text{eu}}$$

*Proof.* 1. Using formula (4.7), it follows immediately that:

$$V_n^{(j),\text{am}} \geq \frac{\pi V_{n+1}^{(j+1),\text{am}} + (1 - \pi) V_{n+1}^{(j),\text{am}}}{1 + r}$$

Which is equivalent to writing:

$$V_n^{\text{am}} \geq \frac{\mathbb{E}^{\mathbb{Q}}[V_{n+1}^{\text{am}} \mid \mathcal{F}_n]}{1 + r}$$

Hence the result.

2. The aim is to show that the strategy consisting in stopping as soon as the value of the option equals the immediate payoff is the best possible one. By (4.7), at each date  $n$ , we have:

$$V_n^{\text{am}} \geq \varphi(S_n)$$

By monotonicity of the expectation and by point 1, we then have for every stopping time  $\tau$ :

$$V_0^{\text{am}} \geq \mathbb{E}^{\mathbb{Q}} \left[ \frac{V_\tau^{\text{am}}}{(1 + r)^\tau} \right] \geq \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_\tau)}{(1 + r)^\tau} \right]$$

Taking the supremum over all stopping times, we obtain:

$$V_0^{\text{am}} \geq \sup_{\tau \text{ s.t.}} \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_\tau)}{(1 + r)^\tau} \right]$$

Define the stopping time  $\tau^*$  as the first instant of equality between the value of the option and the immediate payoff:

$$\tau^* = \min\{n \in \{0, \dots, N\} : V_n^{\text{am}} = \varphi(S_n)\}$$

By definition of the minimum, for every integer  $n$  such that  $0 \leq n < \tau^*$ , we have the strict inequality  $V_n^{\text{am}} > \varphi(S_n)$ . Thus, by (4.7):

$$\forall n < \tau^*, \quad V_n^{\text{am}} = \frac{\pi V_{n+1}^{(j+1),\text{am}} + (1 - \pi) V_{n+1}^{(j),\text{am}}}{1 + r} = \frac{\mathbb{E}^{\mathbb{Q}}[V_{n+1}^{\text{am}} \mid \mathcal{F}_n]}{1 + r}$$

It follows that the sequence of discounted random variables  $\left( \frac{V_n^{\text{am}}}{(1+r)^n} \right)$  is a  $\mathbb{Q}$ -martingale on the time interval  $\llbracket 0, \tau^* \rrbracket$ . Thus:

$$V_0^{\text{am}} = \mathbb{E}^{\mathbb{Q}} \left[ \frac{V_{\tau^*}^{\text{am}}}{(1 + r)^{\tau^*}} \right]$$

Now, by definition of the instant  $\tau^*$ , we have the equality  $V_{\tau^*}^{\text{am}} = \varphi(S_{\tau^*})$ . Substituting this equality into the preceding expression, we obtain:

$$V_0^{\text{am}} = \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_{\tau^*})}{(1 + r)^{\tau^*}} \right]$$

In view of the upper bound found previously, we deduce that:

$$V_0^{\text{am}} = \sup_{\tau \text{ s.t.}} \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_\tau)}{(1 + r)^\tau} \right] = \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_{\tau^*})}{(1 + r)^{\tau^*}} \right]$$

The stopping time  $\tau^*$  attains the maximum of the optimisation problem: it is therefore optimal.

3. For a call,  $\varphi(S_n) = (S_n - K)^+$ . By (3.5), the value  $V_n^{\text{eu}}$  at instant  $n < N$  satisfies:

$$V_n^{\text{eu}} = \mathbb{E}^{\mathbb{Q}} \left[ \frac{(S_N - K)^+}{(1+r)^{N-n}} \middle| \mathcal{F}_n \right] \geq \mathbb{E}^{\mathbb{Q}} \left[ \frac{S_N - K}{(1+r)^{N-n}} \middle| \mathcal{F}_n \right] = S_n - \frac{K}{(1+r)^{N-n}} > S_n - K$$

Moreover, by positivity of the asset under the expectation, we also have  $V_n^{\text{eu}} \geq 0$  for every  $n$ . We deduce that for  $N > n$ :

$$V_n^{\text{eu}} = \mathbb{E}^{\mathbb{Q}} \left[ \frac{(S_N - K)^+}{(1+r)^{N-n}} \middle| \mathcal{F}_n \right] \geq \max(0, S_n - K) = \varphi(S_n)$$

We now show by downward induction on  $n$  that  $V_n^{\text{am}} = V_n^{\text{eu}}$ .

Initialisation ( $n = N$ ): At maturity, both values coincide by definition of the contract:  $V_N^{\text{am}} = V_N^{\text{eu}} = \varphi(S_N)$ .

Inductive step: Suppose the equality holds at instant  $n + 1$ , that is,  $V_{n+1}^{\text{am}} = V_{n+1}^{\text{eu}}$ . At instant  $n$ , the recurrence relation for the American option reads:

$$V_n^{\text{am}} = \max \left( \varphi(S_n), \frac{\mathbb{E}^{\mathbb{Q}}[V_{n+1}^{\text{am}} \mid \mathcal{F}_n]}{1+r} \right)$$

Using the induction hypothesis and the discounted martingale property, the second term of the max is exactly equal to  $V_n^{\text{eu}}$ :

$$V_n^{\text{am}} = \max(\varphi(S_n), V_n^{\text{eu}})$$

Now, we know that  $V_n^{\text{eu}} \geq \varphi(S_n)$ . The max therefore systematically selects the right-hand term, giving:

$$V_n^{\text{am}} = V_n^{\text{eu}}$$

The induction is verified. Thus, the value of the American call is identical to that of the European call at every node of the tree. Immediate exercise is therefore never optimal.  $\square$

This property does not extend to the American put, as we shall now see.

**Definition 4.3.** *The optimal exercise region  $\mathcal{E}$  is the set of all nodes  $(j, n)$  at which the holder of the option has an interest in exercising immediately:*

$$\mathcal{E} := \left\{ (j, n) \mid V_n^{(j), \text{am}} = \varphi(S_n^{(j)}) \quad \text{and} \quad \varphi(S_n^{(j)}) > 0 \right\}. \quad (4.8)$$

*The optimal exercise boundary corresponds to the limiting curve of this region. For a put, it is defined at each date  $n$  by the critical stock price  $S_n^*$  such that:*

$$S_n^* = \max \left\{ S_n^{(j)} \mid (j, n) \in \mathcal{E} \right\}$$

*For a call, it corresponds to the minimal price:*

$$S_n^* = \min \left\{ S_n^{(j)} \mid (j, n) \in \mathcal{E} \right\}$$

Let us examine the origin of the set  $\mathcal{E}$ . In Proposition (4.3), we showed that the optimal stopping time  $\tau^*$  is defined as the first instant at which the value of the option coincides with its immediate payoff:

$$\tau^* = \min\{n \in \{0, \dots, N\} : V_n^{\text{am}} = \varphi(S_n)\}$$

Thus the set  $\mathcal{E}$  represents the investor's decision region. It gathers all nodes  $(j, n)$  for which early exercise is more advantageous than holding the contract, which corresponds to the nodes at which the optimal stopping time  $\tau^*$  is attained.

Mathematically,  $\mathcal{E}$  is defined as the set of nodes of the binomial tree satisfying the optimal exercise conditions. In practice, during the numerical implementation, we translate this set into a Boolean matrix (named `exercise`), where each cell  $(j, i)$  takes the value `True` if the corresponding node belongs to  $\mathcal{E}$ , and `False` otherwise.

The condition  $\varphi(S_n^{(j)}) > 0$  is essential in the implementation. Without it, a node where the immediate payoff is zero and where the continuation value is also zero would be incorrectly classified as an exercise node. The function `snell_envelope` in the notebook guards against this case via the double condition:

```
1 if np.isclose(V[j, i], immediate_gain) and immediate_gain > 0:
2     exercise[j, i] = True
```

Later in the notebook, the function `exercise_boundary` extracts the critical boundary from the Boolean matrix  $\mathcal{E}$ . For each date  $n$ , it identifies the maximal (respectively minimal) index  $j$  among the exercise nodes in column  $n$  for a put (respectively a call). This yields the trajectory of the critical exercise price over time.

**Proposition 4.4.** *We have  $V_0^{\text{am}} \geq V_0^{\text{eu}}$  for any option with non-negative payoff. The inequality is strict for the American put whenever  $K > 0$ .*

*Proof.* We first prove the non-strict inequality. Consider two options with strike  $K$  and maturity  $N$ : one European, the other American. By definition (4.6):

$$V_0^{\text{am}} = \sup_{\tau \text{ s.t.}} \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_\tau)}{(1+r)^\tau} \right]$$

The final maturity  $N$  is a constant; it is the stopping time “wait until the end”. Thus,  $\tau$  may take the value  $N$ . By the property of the supremum, we then have:

$$V_0^{\text{am}} \geq \mathbb{E}^{\mathbb{Q}} \left[ \frac{\varphi(S_N)}{(1+r)^N} \right] = V_0^{\text{eu}}$$

To show that the inequality becomes strict for a put, it suffices to find a node in the tree where early exercise is strictly more profitable than waiting. Consider the extreme scenario in which, at an intermediate date  $n < N$ , the company goes bankrupt or its stock price collapses to  $S_n = 0$ . Since the underlying asset cannot have a negative price, the stock will remain at 0 until maturity ( $S_N = 0$ ).

If the agent holds the American put, they exercise immediately at  $n$ . Their immediate payoff is then maximal and equals:

$$\varphi(S_n) = K - 0 = K$$

If the agent holds the European put, they are forced to wait until maturity  $N$ . The payoff at maturity will also be  $K - 0 = K$ . However, this payoff must be discounted back to the intermediate date  $n$ . Its value at  $n$  is then:

$$V_n^{\text{eu}} = \frac{K}{(1+r)^{N-n}}$$



Since  $r > 0$  and  $N > n$ , we have  $(1 + r)^{N-n} > 1$ . We deduce the strict inequality:

$$K > \frac{K}{(1 + r)^{N-n}} \implies \varphi(S_n) > V_n^{\text{eu}}$$

At date  $n$ , the value of the American option is given by the Snell envelope (4.7), and we then obtain:

$$V_n^{\text{am}} \geq \varphi(S_n) > V_n^{\text{eu}}$$

By backward induction, this strict inequality propagates along the branches of the binomial tree via the Snell envelope. In particular, at  $n = 0$ :

$$V_0^{\text{am}} > V_0^{\text{eu}}$$

□

This inequality is verified numerically in the notebook. A higher price is obtained for the American put than for the European put, while the prices of the American call and the European call are identical, in accordance with Proposition (4.3).

# Chapter 5

## Convergence to continuous time: the Black-Scholes formula

This chapter is the keystone of this dissertation. It answers the following question: what happens when the number of periods  $N$  is taken to infinity while keeping the horizon  $T$  fixed?

The answer to this question brings to light a fundamental result: the price of a European option computed in the CRR binomial model converges to the Black-Scholes-Merton formula. This formula is, as explained in the introduction of this dissertation, the absolute reference for option pricing in continuous time.

This transition from discrete to continuous time is carried out through three successive major phases. First, we will rewrite the expression used to price a European call in a different form. Then, we will adjust the up and down parameters of the tree as a function of the time step. The aim is to ensure that the behaviour of the stock in the discrete model remains consistent when we pass to the continuous time of the Black-Scholes model, that is, when the periods become infinitely short. Next, we will rely on the central limit theorem to carry out the passage to the limit. This approximation will allow us to dispense with the combinatorial computations of the tree by letting the binomial distribution of the number of up-moves converge to the normal distribution, thereby causing the final formula to emerge.

Recall expression (4.5) for pricing a European call with strike  $K$  and maturity  $N$  in the binomial model:

$$C = \frac{1}{(1+r)^N} \sum_{j=0}^N \binom{N}{j} \pi^j (1-\pi)^{N-j} (Sh^j b^{N-j} - K)^+$$

In order to simplify this expression, we introduce the critical threshold  $\eta$  defined by:

$$\eta = \inf \left\{ j \in \mathbb{N} \mid h^j b^{N-j} S - K > 0 \right\}$$

This is the minimal number of up-moves required over the  $N$  periods for the option to be in-the-money, that is, for a strictly positive profit to be guaranteed upon exercise at maturity.

Let us make  $\eta$  explicit. By definition,  $\eta$  is the smallest integer  $j$  satisfying:

$$h^j b^{N-j} S - K > 0 \implies \left( \frac{h}{b} \right)^j b^N S > K \implies \left( \frac{h}{b} \right)^j > \frac{K}{S b^N}$$

Since the natural logarithm is strictly increasing on  $]0, +\infty[$ , we apply it without changing the direction of the inequality:

$$\ln \left[ \left( \frac{h}{b} \right)^j \right] > \ln \left( \frac{K}{Sb^N} \right) \implies j \cdot \ln \left( \frac{h}{b} \right) > \ln \left( \frac{K}{Sb^N} \right)$$

By construction,  $h > b$ , which implies that the ratio  $\frac{h}{b} > 1$ . Thus:

$$j > \frac{\ln \left( \frac{K}{Sb^N} \right)}{\ln \left( \frac{h}{b} \right)}$$

Since  $j$  is an integer, we then have:

$$\eta = \left\lfloor \frac{\ln(K/Sb^N)}{\ln(h/b)} \right\rfloor + 1 \quad (5.1)$$

where  $\lfloor \alpha \rfloor$  is the integer part of the real number  $\alpha$ , that is, the integer satisfying:

$$\lfloor \alpha \rfloor \leq \alpha < \lfloor \alpha \rfloor + 1$$

We can then eliminate the positive part operator  $(\cdot)_+$  by truncating the sum to the relevant states (those for which exercise guarantees a profit). We obtain:

$$C = \frac{1}{(1+r)^N} \sum_{j=\eta}^N \binom{N}{j} \pi^j (1-\pi)^{N-j} (h^j b^{N-j} S - K) \quad (5.2)$$

To formalise this result, we introduce the coefficient  $b(N, j; \pi) = \binom{N}{j} \pi^j (1-\pi)^{N-j}$ , which represents the probability that a random variable following a binomial distribution with parameters  $N$  and  $\pi$  takes the value  $j$ . This allows us to introduce the following notation:

$$B(N, \eta; \pi) = \sum_{j=\eta}^N b(N, j; \pi)$$

Using expression (4.2) for the risk-neutral probability, we directly obtain the following equality:

$$1 - \frac{\pi h}{1+r} = \frac{b - b\pi}{1+r}$$

Distributing the sum in expression (5.2), we obtain:

$$C = \frac{1}{(1+r)^N} \sum_{j=\eta}^N \binom{N}{j} \pi^j (1-\pi)^{N-j} h^j b^{N-j} S - \frac{1}{(1+r)^N} \sum_{j=\eta}^N \binom{N}{j} \pi^j (1-\pi)^{N-j} K$$

Let us examine the first sum. Grouping the terms and using the identity found previously, we have:

$$\begin{aligned} \frac{1}{(1+r)^N} \sum_{j=\eta}^N \binom{N}{j} \pi^j (1-\pi)^{N-j} h^j b^{N-j} S &= S \sum_{j=\eta}^N \binom{N}{j} \left( \frac{\pi h}{1+r} \right)^j \left( \frac{(1-\pi)b}{1+r} \right)^{N-j} \\ &= S \sum_{j=\eta}^N \binom{N}{j} \left( \frac{\pi h}{1+r} \right)^j \left( \frac{b - b\pi}{1+r} \right)^{N-j} \\ &= S \sum_{j=\eta}^N \binom{N}{j} \left( \frac{\pi h}{1+r} \right)^j \left( 1 - \frac{\pi h}{1+r} \right)^{N-j} \\ &= S B \left( N, \eta; \frac{\pi h}{1+r} \right) \end{aligned}$$

Furthermore, the second sum is formalised directly:

$$\begin{aligned} \frac{1}{(1+r)^N} \sum_{j=\eta}^N \binom{N}{j} \pi^j (1-\pi)^{N-j} K &= \frac{K}{(1+r)^N} \sum_{j=\eta}^N \binom{N}{j} \pi^j (1-\pi)^{N-j} \\ &= \frac{K}{(1+r)^N} B(N, \eta; \pi) \end{aligned}$$

These computations lead us to the following proposition:

**Proposition 5.1.** *The price of a European call option in an  $N$ -period binomial model is given by:*

$$C = SB\left(N, \eta; \frac{\pi h}{1+r}\right) - \frac{K}{(1+r)^N} B(N, \eta; \pi) \quad (5.3)$$

By the same reasoning we can establish that the price of a European put option is given by:

$$P = \frac{1}{(1+r)^N} \sum_{j=0}^N \binom{N}{j} \pi^j (1-\pi)^{N-j} (K - h^j b^{N-j} S)^+$$

In order to approach a continuous-time model — the idea being to create time steps that tend to 0 — we wish to pass to the limit in formula (5.3). To do so, we fix a horizon  $T > 0$  and subdivide the interval  $[0, T]$  into  $N \in \mathbb{N}^*$  periods of duration  $dt = T/N$ . Within the Cox, Ross and Rubinstein parametrisation, the up ( $h_N$ ) and down ( $b_N$ ) factors are defined by:

$$h_N = e^{r \cdot dt + \sigma \sqrt{dt}}, \quad b_N = e^{r \cdot dt - \sigma \sqrt{dt}} \quad (5.4)$$

where  $r > 0$  is the global risk-free rate (up to horizon  $T$ ) and  $\sigma > 0$  is the historical volatility of the underlying asset.

Let us examine the global risk-free rate  $r > 0$ . Since the interval  $[0, T]$  is subdivided into  $N$  periods of duration  $dt = T/N$ , it seems natural to suppose that the interest rate for a single period can be defined by:

$$r_N = r \frac{T}{N}$$

After the  $N$  cumulative periods required to reach horizon  $T$ , the total capitalisation factor of the tree is written:

$$(1 + r_N)^N = \left(1 + r \frac{T}{N}\right)^N$$

Passing to the limit, we then obtain:

$$\lim_{N \rightarrow +\infty} \left(1 + \frac{rT}{N}\right)^N = e^{r \cdot T}$$

Of course, each parameter depends on the period length (via the factor  $dt = T/N$ ), which explains the index  $N$ . Let us note that the NAO condition is satisfied by construction. Indeed, to verify the NAO condition, we need  $b_N < e^{r \cdot dt} < h_N$ . Now:

$$b_N = e^{r \cdot dt - \sigma \sqrt{dt}} < e^{r \cdot dt} < e^{r \cdot dt + \sigma \sqrt{dt}} = h_N$$

as soon as  $\sigma > 0$ , which is always the case. Moreover, using expression (4.2), we obtain that the risk-neutral probability associated with this structure takes the form:

$$\pi_N = \frac{e^{r \cdot dt} - b_N}{h_N - b_N} \quad (5.5)$$

We now show why the choice of  $h_N$  and  $b_N$  is judicious.

Let  $J_N \sim \mathcal{B}(N, p)$  be the random variable representing the number of up-moves over the  $N$  periods, where  $p = 1/2$  is the canonical probability. At maturity, using expression (4.1), we define the global log-return of the underlying asset by:

$$\log\left(\frac{S_N}{S}\right) = J_N \log(h_N) + (N - J_N) \log(b_N) = J_N \log\left(\frac{h_N}{b_N}\right) + N \log(b_N) \quad (5.6)$$

Thus, under the canonical probability measure  $\mathbb{P}$  (that is,  $p = 1/2$ ), since  $J_N$  follows a binomial distribution, we readily obtain:

$$\mathbb{E}^{\mathbb{P}}[J_N] = N/2 \quad \text{and} \quad \text{Var}^{\mathbb{P}}(J_N) = N/4$$

Let us now compute the expectation (denoted  $N\mu_N$ ) and the variance (denoted  $N\sigma_N^2$ ) of the log-return  $\ln\left(\frac{S_N}{S}\right)$ . First, let us simplify expression (5.6):

$$\ln(b_N) = \ln\left(e^{r \cdot dt - \sigma \sqrt{dt}}\right) = r \cdot dt - \sigma \sqrt{dt}$$

Moreover:

$$\ln\left(\frac{h_N}{b_N}\right) = \ln(h_N) - \ln(b_N) = (r \cdot dt + \sigma \sqrt{dt}) - (r \cdot dt - \sigma \sqrt{dt}) = 2\sigma \sqrt{dt}$$

Substituting these simplified blocks, our log-return variable takes the simpler form:

$$\ln\left(\frac{S_N}{S}\right) = J_N (2\sigma \sqrt{dt}) + N (r \cdot dt - \sigma \sqrt{dt}) \quad (5.7)$$

Thus, by linearity of the expectation, we obtain:

$$N\mu_N = \mathbb{E}^{\mathbb{P}} \left[ \ln\left(\frac{S_N}{S}\right) \right] = \mathbb{E}^{\mathbb{P}}[J_N] \cdot (2\sigma \sqrt{dt}) + N (r \cdot dt - \sigma \sqrt{dt})$$

Replacing  $\mathbb{E}^{\mathbb{P}}[J_N]$  by  $\frac{N}{2}$ :

$$N\mu_N = \left(\frac{N}{2}\right) \cdot 2\sigma \sqrt{dt} + N \cdot r \cdot dt - N \cdot \sigma \sqrt{dt}$$

Simplifying, we obtain:

$$N\mu_N = N \cdot r \cdot \left(\frac{T}{N}\right) = r \cdot T$$

The expectation of the log-return is therefore  $rT$ . We now compute the variance of the log-return. By the properties of the variance and using expression (5.7), we obtain:

$$N\sigma_N^2 = \text{Var}^{\mathbb{P}} \left[ \ln\left(\frac{S_N}{S}\right) \right] = (2\sigma \sqrt{dt})^2 \cdot \text{Var}^{\mathbb{P}}(J_N)$$

Replacing  $\text{Var}^{\mathbb{P}}(J_N)$  by  $\frac{N}{4}$  and simplifying:

$$N\sigma_N^2 = \sigma^2 \cdot dt \cdot N = \sigma^2 \cdot \left(\frac{T}{N}\right) \cdot N = \sigma^2 \cdot T$$

The variance of the log-return is therefore  $\sigma^2 T$ . It is then very important to note that these parameters are independent of  $N$ . This allows us to deduce that, regardless of the subdivision, the expectation and variance of the log-return remain  $rT$  and  $\sigma^2 T$  respectively, which justifies the relevance of the choice of parameters  $h_N$  and  $b_N$ .

The fundamental objective of the transition to continuous time is to ensure that our discrete model converges, at horizon  $T$ , to the true continuous-time stock price, that is, that  $S_N$  converges to  $S_T$  in the following sense:

$$\lim_{N \rightarrow +\infty} \mathbb{E} \left[ \ln \left( \frac{S_N}{S} \right) \right] = \mathbb{E} \left[ \ln \left( \frac{S_T}{S} \right) \right] \quad \text{and} \quad \lim_{N \rightarrow +\infty} \text{Var} \left[ \ln \left( \frac{S_N}{S} \right) \right] = \text{Var} \left[ \ln \left( \frac{S_T}{S} \right) \right] \quad (5.8)$$

Or equivalently:

$$\lim_{N \rightarrow +\infty} N\mu_N = \mu T \quad \text{and} \quad \lim_{N \rightarrow +\infty} N\sigma_N^2 = \sigma^2 T$$

where  $\mu$  and  $\sigma^2$  represent respectively the “instantaneous” expectation and variance<sup>1</sup> of the log-return of the continuous-time model. Performing the computations above with conditions (5.8), it would then have been natural to take the following parameters:

$$h_N = e^{\mu \cdot dt + \sigma \sqrt{dt}}, \quad b_N = e^{\mu \cdot dt - \sigma \sqrt{dt}}$$

However, there would be a major problem: the parameters would be impossible to compute precisely. Indeed, the average return of the stock (under  $\mathbb{P}$ ) depends on the future, on the expectations of agents, on the risk premium demanded by investors, and so on. It is very difficult to estimate. It is at this point that one understands the importance of pricing under the risk-neutral measure  $\mathbb{Q}$ . In the risk-neutral “world”, we can dispense entirely with agents’ risk preferences and, consequently, with the historical expected return  $\mu$ , which is replaced by the risk-free rate  $r$ . This substitution is extremely convenient because, unlike  $\mu$ , the rate  $r$  is a fixed and observable quantity. Furthermore, this simplification is crucial for the implementation in our notebook, as it allows the function `parameters` to yield economically viable results.

The framework being established, we will now study the behaviour of (5.3) as  $N \rightarrow +\infty$ . First, the up and down parameters depend on  $N$ , so the critical threshold defined by (5.1) also depends on  $N$ . We denote it:

$$\eta_N = \left\lfloor \frac{\ln(K/Sb_N^N)}{\ln(h_N/b_N)} \right\rfloor + 1 \quad (5.9)$$

We then write (5.3) in the following form:

$$C = S \cdot B(N, \eta_N; \frac{\pi_N h_N}{1 + r_N}) - \frac{K}{(1 + r_N)^N} \cdot B(N, \eta_N; \pi_N)$$

---

<sup>1</sup>In continuous time, the “instantaneous” expectation and volatility measure respectively the expected average return and the risk of an asset over an infinitely short period. They are rescaled to the overall period  $T$ .

Let us detail the computations for  $B(N, \eta_N; \pi_N)$ . We consider  $Y_N = \sum_{i=1}^N X_{i,N}$  the sum of  $N$  independent Bernoulli random variables with parameter  $\pi_N$  ( $\forall i, N, X_{i,n} \sim \mathcal{B}(\pi_N)$ ). We begin by noting the following identity:

$$B(N, \eta_N; \pi_N) = 1 - \mathbb{P}(Y_N < \eta_N)$$

Indeed, by definition, we have:

$$B(N, \eta_N; \pi_N) = \sum_{j=\eta_N}^N d(N, j; \pi_N)$$

Now,  $d(N, j; \pi_N) = \binom{N}{j} \pi_N^j (1 - \pi_N)^{N-j} = \mathbb{P}(Y_N = j)$ , so we can rewrite this sum in the form:

$$B(N, \eta_N; \pi_N) = \sum_{j=\eta_N}^N \mathbb{P}(Y_N = j)$$

Since the events  $\{Y_N = j\}$  are pairwise disjoint for each value of  $j$ , we have:

$$\sum_{j=\eta_N}^N \mathbb{P}(Y_N = j) = \mathbb{P}\left(\bigcup_{j=\eta_N}^N \{Y_N = j\}\right)$$

Moreover, since  $Y_N \sim \mathcal{B}(\pi_N)$ , we have  $\text{Supp}(Y_N) = \{0, 1, \dots, N\}$  and thus:

$$\bigcup_{j=\eta_N}^N \{Y_N = j\} = \{Y_N \geq \eta_N\}$$

Finally, taking the complement, we obtain:

$$B(N, \eta_N; \pi_N) = \mathbb{P}(Y_N \geq \eta_N) = 1 - \mathbb{P}(Y_N < \eta_N) \quad (5.10)$$

In order to study the convergence of  $B(N, \eta_N; \pi_N)$ , we standardise the random variable  $Y_N$ :

$$\mathbb{P}(Y_N < \eta_N) = \mathbb{P}\left(\frac{Y_N - N\pi_N}{\sqrt{N\pi_N(1 - \pi_N)}} < \frac{\eta_N - N\pi_N}{\sqrt{N\pi_N(1 - \pi_N)}}\right) \quad (5.11)$$

since  $\mathbb{E}^\mathbb{P}[Y_N] = N\pi_N$  and  $\text{Var}(Y_N) = N\pi_N(1 - \pi_N)$ .

We will now attempt to express the critical threshold  $\eta_N$  differently. Applying the logarithm to  $\frac{h_N}{b_N} = e^{2\sigma\sqrt{dT}}$  and to  $b_N^N$ , we obtain:

$$\ln(h_N/b_N) = 2\sigma\sqrt{\frac{T}{N}} \quad \text{and} \quad N \ln b_N = rT - \sigma\sqrt{TN}$$

Thus:

$$\begin{aligned} \frac{\ln(K/Sb_N^N)}{\ln(h_N/b_N)} &= \frac{\ln(K/S) - rT + \sigma\sqrt{TN}}{2\sigma\sqrt{\frac{T}{N}}} \\ &= \frac{\sqrt{N}}{2\sigma\sqrt{T}} (\ln(K/S) - rT + \sigma\sqrt{TN}) \\ &= \frac{\sigma\sqrt{TN}}{2\sigma\sqrt{T}} + \frac{\ln(K/S) - rT}{2\sigma\sqrt{T}} \sqrt{N} \\ &= \frac{N}{2} + \frac{\ln(K/S) - rT}{2\sigma\sqrt{T}} \sqrt{N} \end{aligned}$$

Let us denote  $\alpha_N = \frac{N}{2} + \frac{\ln(K/S) - rT}{2\sigma\sqrt{T}}\sqrt{N}$ . By expression (5.9), we have:

$$\eta_N = \lfloor \alpha_N \rfloor + 1$$

We then write:

$$\alpha_N = \lfloor \alpha_N \rfloor + r_N, \quad 0 \leq r_N < 1$$

where  $r_N$  represents the fractional part of  $\alpha_N$ . Thus:

$$\begin{aligned} \eta_N - \alpha_N &= (\lfloor \alpha_N \rfloor + 1) - (\lfloor \alpha_N \rfloor + r_N) \\ &= 1 - r_N \end{aligned}$$

Since  $0 \leq r_N < 1$ , we deduce:

$$0 < 1 - r_N \leq 1$$

Thus,  $\eta_N - \alpha_N = 1 - r_N$  is bounded. We then obtain:

$$\eta_N - \alpha_N = o(\sqrt{N})$$

We conclude:

$$\eta_N = \frac{N}{2} + \frac{\ln(K/S) - rT}{2\sigma\sqrt{T}}\sqrt{N} + o(\sqrt{N}) \quad (5.12)$$

Let us proceed likewise with the risk-neutral probability defined by (5.5). Substituting the expressions for  $h_N$  and  $b_N$ , we have:

$$\pi_N = \frac{e^{r\frac{T}{N}} \left( 1 - e^{-\sigma\sqrt{\frac{T}{N}}} \right)}{e^{r\frac{T}{N}} \left( e^{\sigma\sqrt{\frac{T}{N}}} - e^{-\sigma\sqrt{\frac{T}{N}}} \right)} = \frac{1 - e^{-\sigma\sqrt{\frac{T}{N}}}}{e^{\sigma\sqrt{\frac{T}{N}}} - e^{-\sigma\sqrt{\frac{T}{N}}}}$$

Now, since  $\sqrt{dt} \rightarrow 0$ , by Taylor expansion of the exponential, we have:

$$e^{-\sigma\sqrt{dt}} = 1 - \sigma\sqrt{dt} + \frac{\sigma^2}{2}dt + o(dt)$$

Therefore:

$$1 - e^{-\sigma\sqrt{dt}} = \sigma\sqrt{dt} - \frac{\sigma^2}{2}dt + o(dt)$$

And since  $e^{\sigma\sqrt{dt}} - e^{-\sigma\sqrt{dt}} = 2\sigma\sqrt{dt} + O(dt^{3/2})$ , we can write  $\pi_N$  in the form:

$$\pi_N = \frac{\sigma\sqrt{dt} - \frac{\sigma^2}{2}dt + o(dt)}{2\sigma\sqrt{dt} + O(dt^{3/2})}$$

Simplifying:

$$\begin{aligned} \pi_N &= \frac{1 - \frac{\sigma}{2}\sqrt{dt} + o(\sqrt{dt})}{2 + O(dt)} = \frac{1}{2} \left( 1 - \frac{\sigma}{2}\sqrt{dt} + o(\sqrt{dt}) \right) \\ &= \frac{1}{2} - \frac{\sigma}{4}\sqrt{dt} + o(\sqrt{dt}) \end{aligned}$$

Finally, we obtain the following expansion:

$$\pi_N = \frac{1}{2} - \frac{\sigma\sqrt{T}}{4\sqrt{N}} + o\left(\frac{1}{\sqrt{N}}\right) \quad (5.13)$$



Substituting the results (5.12) and (5.13), we obtain:

$$\begin{aligned}\eta_N - N\pi_N &= \left( \frac{\ln(K/S) - rT}{2\sigma\sqrt{T}} + \frac{\sigma\sqrt{T}}{4} \right) \sqrt{N} + o(\sqrt{N}) \\ &= \frac{\ln(K/S) - rT + \frac{1}{2}\sigma^2 T}{2\sigma\sqrt{T}} \sqrt{N} + o(\sqrt{N}) \\ &= \frac{\ln(K/S_0) - \left(r - \frac{1}{2}\sigma^2\right) T}{2\sigma\sqrt{T}} \sqrt{N} + o(\sqrt{N})\end{aligned}$$

Moreover, since  $\pi_N \rightarrow \frac{1}{2}$ , it follows that:

$$\pi_N(1 - \pi_N) = \frac{1}{4} + o(1)$$

Thus, multiplying by  $\sqrt{N}$ :

$$\sqrt{N\pi_N(1 - \pi_N)} = \frac{\sqrt{N}}{2} + o(\sqrt{N})$$

We can then assemble the following quotient in order to study its limit:

$$\frac{\eta_N - N\pi_N}{\sqrt{N\pi_N(1 - \pi_N)}} = \frac{\frac{\ln(K/S) - \left(r - \frac{1}{2}\sigma^2\right) T}{2\sigma\sqrt{T}} \sqrt{N} + o(\sqrt{N})}{\frac{\sqrt{N}}{2} + o(\sqrt{N})}$$

Dividing through by  $\sqrt{N}$ , the  $o(\sqrt{N})$  terms become  $o(1)$  and vanish in the limit:

$$\lim_{N \rightarrow \infty} \frac{\eta_N - N\pi_N}{\sqrt{N\pi_N(1 - \pi_N)}} = \frac{\ln(K/S) - \left(r - \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}} \quad (5.14)$$

At this stage, we would like to apply the central limit theorem. Unfortunately, this is not possible, since the distribution of the variables  $X_{i,N}$  for  $i \in \{1, \dots, N\}$  depends on  $N$  through the parameter  $\pi$ . We will therefore circumvent this difficulty via characteristic functions.

Let  $\phi_X(t) = \mathbb{E}[e^{itX}]$  be the characteristic function of a real random variable  $X$ . If  $X$  is centred and standardised, we have:

$$\begin{aligned}\phi_X(t) &= 1 + it\mathbb{E}[X] - \frac{t^2}{2}\mathbb{E}[X^2] + o(t^2) \\ &= 1 - \frac{t^2}{2}\text{Var}(X) + o(t^2)\end{aligned}$$

since  $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathbb{E}[X^2]$ .

Since the variables  $X_{i,N}$  are independent, we have:

$$\mathbb{E} \left[ \exp \left( it \frac{Y_N - N\pi_N}{\sqrt{N\pi_N(1 - \pi_N)}} \right) \right] = \left( \mathbb{E} \left[ \exp \left( it \frac{X_{1,N} - \pi_N}{\sqrt{N\pi_N(1 - \pi_N)}} \right) \right] \right)^N$$

The variables  $\frac{X_{1,N} - \pi_N}{\sqrt{N\pi_N(1-\pi_N)}}$  are centred, with variance  $1/N$ , from which:

$$\mathbb{E} \left[ \exp \left( it \frac{X_{1,N} - \pi_N}{\sqrt{N\pi_N(1-\pi_N)}} \right) \right] = 1 - \frac{t^2}{2N} + o\left(\frac{1}{N}\right)$$

Passing to exponential form, we can write:

$$\left( 1 - \frac{t^2}{2N} + o\left(\frac{1}{N}\right) \right)^N = \exp \left[ N \ln \left( 1 - \frac{t^2}{2N} + o\left(\frac{1}{N}\right) \right) \right]$$

Since  $\lim_{N \rightarrow \infty} \left( -\frac{t^2}{2N} + o\left(\frac{1}{N}\right) \right) = 0$ , we can use the Taylor expansion  $\ln(1+u) = u + o(u)$ , giving:

$$N \ln \left( 1 - \frac{t^2}{2N} + o\left(\frac{1}{N}\right) \right) = -\frac{t^2}{2} + o(1)$$

By continuity of the exponential function:

$$\lim_{N \rightarrow \infty} \left( 1 - \frac{t^2}{2N} + o\left(\frac{1}{N}\right) \right)^N = \exp \left( -\frac{t^2}{2} \right)$$

We recognise the characteristic function of the standard normal distribution ( $t \mapsto \exp(-\frac{t^2}{2})$ ). Thus, by Lévy's continuity theorem:

$$\frac{Y_N - N\pi_N}{\sqrt{N\pi_N(1-\pi_N)}} \xrightarrow[N \rightarrow \infty]{\mathcal{L}} \mathcal{N}(0, 1) \quad (5.15)$$

Denote by

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{u^2}{2}} du$$

the cumulative distribution function of the standard normal distribution  $\mathcal{N}(0, 1)$ . Since  $\phi$  is continuous, convergence in distribution implies uniform convergence of the cumulative distribution function on  $\mathbb{R}$ . This justifies the following passage to the limit in (5.11):

$$\mathbb{P} \left( \frac{Y_N - N\pi_N}{\sqrt{N\pi_N(1-\pi_N)}} < \frac{\eta_N - N\pi_N}{\sqrt{N\pi_N(1-\pi_N)}} \right) \xrightarrow[N \rightarrow \infty]{} \phi \left( \frac{\ln(K/S) + \left(r - \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}} \right) \quad (5.16)$$

Gathering results (5.10), (5.14), (5.15) and (5.16), we obtain:

$$B(N, \eta_N; \pi_N) \xrightarrow[N \rightarrow \infty]{} 1 - \phi \left( \frac{\ln(K/S) - \left(r - \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}} \right)$$

Exploiting the symmetry property of the normal distribution ( $1 - \phi(x) = \phi(-x)$ ), the expression simplifies to:

$$B(N, \eta_N; \pi_N) \xrightarrow[N \rightarrow \infty]{} \phi \left( \frac{\ln(S/K) + \left(r - \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}} \right)$$

The same methods apply to study the second term of (5.3),  $B(N, \eta_N; \frac{\pi_N h_N}{1+r_N})$ . Thus, applying these results to (5.3), we arrive at the following theorem.

**Theorem 5.1.** *As  $N \rightarrow \infty$ , with the parameters (5.4), the price  $C$  of a European call converges to:*

$$S\phi(d) - Ke^{-rT}\phi(d - \sigma\sqrt{T}) \quad (5.17)$$

with:

$$d = \frac{\ln(S/K) + rT}{\sigma\sqrt{T}} + \frac{1}{2}\sigma\sqrt{T}$$

We can observe that this limiting price does not depend on the parameter  $\mu$  representing the expected rate of return of the asset. This may seem contradictory at first, since it is natural to think that an investor would be willing to pay more for a call if the stock has a strong upward tendency (that is, a high  $\mu$ ). Yet the formula shows that two stocks with the same initial price  $S$  and the same volatility  $\sigma$ , but with entirely different trajectories (one soaring due to a high  $\mu$ , the other stagnating, for example), will yield exactly the same price for the call.

This is in fact far from trivial. Indeed, all our computations have been carried out under the risk-neutral measure. Thus, everything proceeds as if investors were indifferent to risk: the parameter  $\mu$  is purely and simply “replaced” by the risk-free rate  $r$ . The price is then determined solely by  $r$  and the volatility  $\sigma$ .

Furthermore, as explained previously, this property is very convenient for implementation. As a reminder, the historical drift  $\mu$  is very difficult to estimate precisely, whereas the risk-free rate  $r$  and the volatility  $\sigma$  are directly observable quantities. It is these conveniences that make the formula computationally tractable, and that allowed us to implement formula (5.17) in the function `black_scholes`:

```
1 def black_scholes(S, K, r, T, sigma):
2     d = (np.log(S/K) + r*T) / (sigma * np.sqrt(T)) + sigma * np.sqrt(T)
3     C = S * norm.cdf(d) - K * np.exp(-r*T) * norm.cdf(d - sigma * np.
4         sqrt(T))
5     return C
```

By the same reasoning we obtain that the price of a put  $P$  converges to:

$$Ke^{-rT}\phi(\delta + \sigma\sqrt{T}) - S\phi(\delta)$$

with:

$$\delta = \frac{\ln(K/S) - rT}{\sigma\sqrt{T}} - \frac{1}{2}\sigma\sqrt{T}$$

This allows us to deduce the put-call parity formula:

$$C = P + S - e^{-rT}K \quad (5.18)$$

Formula (5.17) is known as the Black-Scholes formula and is regarded as one of the most important discoveries in the history of modern finance. This major contribution earned its authors, Robert Merton and Myron Scholes, the Nobel Prize in Economics in 1997, Fischer Black having unfortunately passed away before he could receive it.

Before its publication in 1973, valuing an option was often a matter of guesswork or intuition, in the absence of a universal method for determining a “fair price”. Black, Scholes and Merton provided a mathematical solution by showing that it is possible to compute the price of an option from entirely observable variables (the stock price, the

strike price, the time remaining, the risk-free rate, and the volatility).

Moreover, the formula proves that the risk associated with an option can be neutralised. For financial institutions, this changed everything. Any bank could henceforth sell an option to a client and mathematically “hedge” its own risk by continuously adjusting its portfolio.

Furthermore, this formula triggered the explosion of the derivatives market. As soon as investors had a reliable tool for setting prices and managing risk, confidence in the markets surged. The *Chicago Board Options Exchange*<sup>2</sup> opened its doors just one month before the publication of the formula. Thanks to it, the derivatives market grew from a small circle of speculators into a global market worth hundreds of trillions of dollars today.

Today, the logic of the formula is applied to what are known as “real options”. Companies use it, for example, to decide whether to wait before launching a project (a new drug, for instance), or to value stock options<sup>3</sup> granted to employees.

However, it is important to note that this model is not perfect. It rests on the assumption that prices evolve smoothly and without sudden jumps. History has unfortunately revealed its limitations during unforeseen crashes, as illustrated by the collapse of the hedge fund LTCM in 1998, a reminder that a mathematical formula remains a simplified model of reality.

A natural question is to ask at what rate the price of a call defined by (4.5) converges to the Black-Scholes formula (5.17).

The function `convergence` allows us to answer this question by plotting the evolution of the binomial price  $C_N$  for  $N$  varying from 1 to  $N_{\max}$ , in order to compare it visually with the theoretical continuous value ( $C_{BS}$ ):

```
1 def convergence(S, K, r, T, sigma, N_max):
2     binomial_prices = []
3     for i in range(1, N_max+1):
4         dt, h, b, pi = parameters(T, r, sigma, i)
5         M = price_tree(S, h, b, i)
6         price, V = option_value2(M, K, r, dt, pi, i, "call")
7         binomial_prices.append(price)
8     plt.plot(range(1, N_max+1), binomial_prices, color="blue", label="
    Binomial")
9     plt.plot(range(1, N_max+1), [black_scholes(S, K, r, T, sigma)] *
    N_max,
10             color="red", label="Black-Scholes")
11     plt.xlabel("Number of steps N")
12     plt.ylabel("Call price")
13     plt.title("Convergence of the binomial model towards Black-Scholes")
14     plt.legend()
15     plt.show()
```

---

<sup>2</sup>The CBOE (now known as *Cboe Global Markets*) is historically the very first exchange in the world dedicated exclusively to the trading of negotiable options contracts.

<sup>3</sup>A stock option is a right granted by a company to its employees to purchase shares of the company at a predetermined price. Its purpose is to retain employees and motivate them to increase the value of the company.

For  $S = K = 100$ ,  $T = 1$ ,  $r = 5\%$ ,  $\sigma = 20\%$  and  $N_{\max} = 100$ , the limiting price computed is  $C_{BS} \approx 10.45$  €.

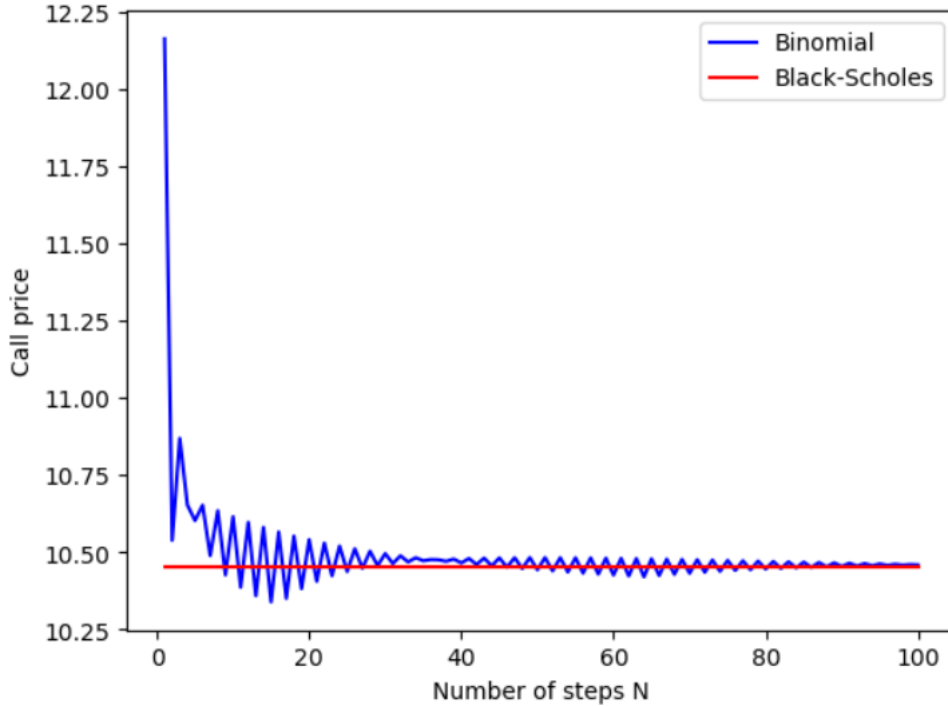


Figure 5.1: Convergence of the binomial model towards the Black-Scholes model ( $N_{\max} = 100$ )

Figure (5.1) confirms that the greater the number of steps  $N$ , the closer the tree price approaches the theoretical Black-Scholes price.

Visually, the curve begins to stabilise around the target value from  $N = 30$  onwards. Moreover, in the reasoning used to arrive at the Black-Scholes formula, we could have applied the central limit theorem<sup>4</sup>. We can then deduce that the convergence is of order  $O(1/\sqrt{N})$ , which corresponds to a rather slow rate of convergence.

Furthermore, oscillations around the theoretical price are observed. When the number of steps  $N$  is even, the strike price  $K$  coincides exactly with a node of the tree. Since the node where  $S = K$  is fixed and equal to 0, the discrete model ignores continuous infinitesimal variations. Conversely, when  $N$  is odd, no node coincides with  $K$ . During the recursive backward induction on the tree, the expectation of the gains performs a “smoothing” between adjacent nodes, which overestimates the price.

Consider an example where  $K = 100$  and  $N = 1$ . At maturity, the only possible prices for the underlying are located on either side of the strike price (for example  $S_1 = 105$  and  $S_1 = 95$ ). The final payoffs are exact (5 and 0) and the backward induction step at the initial date  $t = 0$  (where  $S_0 = 100$ ) computes a weighted expectation between these two states according to formula (4.3). The algorithm then behaves as if the payoff between 95 and 105 evolved along a continuous straight line. Yet, if the price of the underlying is below  $K = 100$ , it is worth nothing. We have therefore counted nodes where the option theoretically yields nothing (such as 96, 97.2, 98.673, etc.), which naturally gives rise to an overestimation of the price.

<sup>4</sup>More precisely, the Lindeberg theorem, which constitutes a generalisation of the central limit theorem.

Let us now consider the case where  $N = 2$ . Since the tree is recombining, we can now return to the origin where the price of the underlying equals the strike ( $S_2 = 100$ ). At this node, the payoff is zero ( $\max(100 - 100, 0) = 0$ ). However, in the continuous Black-Scholes model, an asset worth 100 at a given instant retains a probability of oscillating infinitesimally just above the strike before maturity (for example, 100.01 or 99.98). Since up-moves generate a gain and down-moves are floored at zero, these micro-variations do not cancel out and contribute a positive value that the binomial tree (for even  $N$ ) entirely suppresses by fixing the node at zero. Thus, the discrete model underestimates the price.

In summary, this chapter has shown that the binomial model is theoretically consistent and that it does indeed converge to the Black-Scholes formula. Now that the theory has been validated, it is time to move on to practical implementation. The next chapter will therefore be dedicated to the numerical implementation of these models, with the concrete objective of applying them to a *Crédit Agricole* warrant<sup>56</sup>.

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<sup>5</sup>A warrant functions like an option, with the difference that it is issued in limited quantity, does not allow short selling, and is traded directly on the stock exchange like a share. Furthermore, it is issued by a financial institution.

<sup>6</sup>Cf. Chapter 7.

# Chapter 6

## Architecture of the Python implementation

The fundamental idea of the computer program associated with this dissertation is to put into practice the notions developed previously in order to price financial options. To this end, we have implemented several functions, each corresponding to a result proved earlier.

The purpose of this chapter is to describe this correspondence between theory and implementation, explaining not only what each function does, but why it is constructed in that way; that is, which theoretical result is being implemented and what algorithmic choices follow from it.

The structure of this chapter follows the logical organisation of the theoretical part. The first section is devoted to model calibration, the second addresses the construction of the binomial tree, and the third presents the option pricing algorithms.

The implementation was carried out in Python 3.8+, relying on the standard scientific ecosystem: **numpy** for matrix manipulations, **scipy.stats** for the cumulative distribution function of the normal distribution, and **pandas** for the processing of market parameters.

Unless otherwise stated, all numerical applications and graphical verifications in this chapter will be carried out using the following reference parameters:  $S = 100$ ,  $K = 100$ ,  $T = 1$ ,  $r = 5\%$  and  $\sigma = 20\%$ .

The implementation begins with the function **parameters**. This is the transcription of the CRR parametrisation developed in Chapter 5. It takes as input the continuous-time model parameters  $(T, r, \sigma, N)$  and returns the discrete parameters  $(dt, h, b, \pi)$ .

```
1 def parameters(T, r, sigma, N):
2     dt = T / N
3     h = np.exp(r*dt + sigma*np.sqrt(dt))
4     b = np.exp(r*dt - sigma*np.sqrt(dt))
5     pi = (np.exp(r*dt) - b) / (h - b)
6     return dt, h, b, pi
```

Let us recall what each parameter corresponds to. The variable **dt** defines the time step by subdividing the horizon into  $N$  regular intervals. The variables **h** and **b** compute the up and down multiplicative factors  $h_N = e^{r \cdot dt + \sigma \sqrt{dt}}$  and  $b_N = e^{r \cdot dt - \sigma \sqrt{dt}}$  as defined by formula (5.4). Finally, the variable **pi** determines the risk-neutral probability  $\pi_N = (e^{r \cdot dt} - b_N) / (h_N - b_N)$ , following expression (5.5).

Let us also recall that the NAO condition, characterised by the inequality  $b_N < e^{r \cdot dt} <$

$h_N$ , is satisfied by construction. It is therefore unnecessary to add a further test within the code to verify it.

We will now construct the binomial tree through the function `price_tree`. The recombining structure of the binomial model is encoded in a lower triangular matrix  $M$  defined by relation (4.1):  $M_{j,n} = S \cdot h^j \cdot b^{n-j}$  for  $0 \leq j \leq n \leq N$ .

```
1 def price_tree(S, h, b, N):
2     M = np.zeros((N+1, N+1))
3     for i in range(N+1):
4         for j in range(i+1):
5             M[j, i] = S * (h**j) * (b**(i-j))
6     return M
```

The tree is represented by an  $(N + 1) \times (N + 1)$  matrix in which only the lower triangular part is filled. Visually, the idea rests on Figure (4.2). Thus, the column index  $i$  represents the date  $t = i$ , and the row index  $j$  represents the number of up-moves that have occurred since  $t = 0$ . In this configuration, the first row  $M[0, i]$  represents the path consisting entirely of down-moves, forming the lowest trajectory of the tree. At the other extreme, the diagonal  $M[i, i]$  gathers the prices after an uninterrupted succession of up-moves. All other elements  $M[j, i]$  (with  $0 < j < i$ ) represent intermediate states in which the asset undergoes  $j$  up-moves and  $i - j$  down-moves, in any order.

Let us recall that this last property — the recombination of the tree — is particularly convenient, indeed necessary, for our computations. It reduces the complexity from  $\mathcal{O}(2^N)$  distinct nodes to  $\mathcal{O}(N^2)$ .

The function `payoff` constitutes the algorithmic translation of the option's payoff profile at maturity. It takes as input the price of the underlying, the strike price, and the nature of the contract, and returns the corresponding gain:

```
1 def payoff(S, K, option_type):
2     if option_type == "call":
3         return max(S - K, 0)
4     elif option_type == "put":
5         return max(K - S, 0)
```

The formulas implemented are  $\varphi(S) = (S - K)^+$  for a call, and  $\varphi(S) = (K - S)^+$  for a put. This function is invoked in two ways within the program: it is called at each terminal node for the initialisation of the recurrence, and at each intermediate node for the American option.

As we saw in Chapter 4, we have two methods for pricing a European option: via a closed-form formula and via backward induction on the tree. The function `option_value1` implements the closed-form formula (4.5):

$$C = e^{-rT} \sum_{j=0}^N \binom{N}{j} \pi^j (1 - \pi)^{N-j} \varphi(S \cdot h^j \cdot b^{N-j}).$$

Note that the discounting coefficient  $\frac{1}{(1+r)^N}$  is replaced by  $e^{-rT}$ <sup>1</sup>.

```
1 def option_value1(S, K, r, T, h, b, pi, N, option_type):
2     total = 0
```

<sup>1</sup>Cf. Chapter 5 for a detailed justification of this passage to the limit.



```

3     for j in range(0, N+1):
4         S_node = (h**j) * (b**(N-j)) * S
5         total += math.comb(N, j) * (pi**j) * ((1-pi)**(N-j)) * payoff(
        S_node, K, option_type)
6     return np.exp(-r*T) * total

```

The loop computes the binomial sum by traversing all states  $j \in \{0, \dots, N\}$  of the tree at date  $N$ . The discount factor is then applied outside the loop. This approach proves particularly efficient for European options since it requires only a single pass over the  $N + 1$  terminal nodes, thereby maintaining a linear complexity of  $\mathcal{O}(N)$ .

To go beyond the closed-form formula and enable in particular the pricing of American options, it is necessary to implement the backward induction algorithm. To this end, the function `option_value2` implements the dynamic programming principle based on equation (4.3) from Chapter 4:

$$V_n^{(j)} = \frac{\pi V_{n+1}^{(j+1)} + (1 - \pi) V_{n+1}^{(j)}}{1 + r}$$

Moreover, in the case of an American option, this discounted value is compared at each node with the immediate payoff, in accordance with relation (4.7).

```

1 def option_value2(M, K, r, dt, pi, N, option_type, american=False):
2     V = np.zeros(np.shape(M))
3     #Initialisation at maturity
4     for j in range(0, N+1):
5         V[j, N] = payoff(M[j, N], K, option_type)
6
7     #Recursive backward induction on the tree
8     for i in range(N-1, -1, -1):
9         for j in range(0, i+1):
10             discounted_value = np.exp(-r*dt) * ((pi * V[j+1, i+1]) + (1-
11 pi) * V[j, i+1])
12             if not american:
13                 V[j, i] = discounted_value
14             else:
15                 V[j, i] = max(payoff(M[j, i], K, option_type),
16 discounted_value)
17     return V[0, 0], V

```

The code follows a reverse chronological logic that begins with an initialisation phase. At maturity (column  $N$ ), the matrix  $V$  is filled with the terminal payoff values  $\varphi(M_{j,N})$  for each state  $j \in \{0, \dots, N\}$ . This is the equivalent of the terminal condition  $V_N^{(j)} = \varphi(S_N^{(j)})$ , in accordance with relation (4.4).

Once maturity has been initialised, the program proceeds with the recursive backward induction on the tree. The double `for` loop traverses the columns from right to left, from date  $n = N - 1$  back to the origin  $n = 0$ , while the index  $j$  traverses the rows of column  $i$ . Again, the discount factor is  $e^{-r \cdot dt}$  rather than  $(1 + r)^{-1}$ , consistent with the continuous-time parametrisation.

This structure entails an index correspondence between a node and its successors, which we have already discussed in Chapter 4. The reader may refer to the detailed explanations accompanying Figure (4.3).

When an American-type option is selected, the instruction `max(payoff(...), discounted_value)` introduces the possibility of early exercise. At each node, the algorithm compares the immediate payoff with the continuation value if the option contract is held. This mechanism

materialises the concept of the Snell envelope introduced with formula (4.7) in Chapter 4.

Finally, the function returns the pair of variables  $(V[0,0], V)$ . The first term provides the option price computed at the initial date  $t = 0$ , while the matrix  $V$  preserves the history of nodal values, which is useful for visualisations and for detecting the optimal exercise boundary.

After implementing the computation and backpropagation functions, the function `visualize_tree` allows us to display the binomial tree (cf. Figure 6.1). This display is generated using our reference values for a call (the colour of the nodes is inverted for a put).

This graph makes it possible to consolidate a large amount of data. At each node, the stock price is displayed in black, while the corresponding option value appears in blue. The trajectories are materialised by coloured arrows, where green arrows represent up-moves and red arrows represent down-moves. Finally, the node colours indicate the position of the underlying relative to the strike  $K$ : green when  $S_t > K$  (call in-the-money), red when  $S_t < K$  (out-of-the-money), and black when  $S_t = K$  (at-the-money). This convention facilitates the reading of the tree and highlights the states that are favourable or unfavourable to the exercise of the option at maturity.

It should however be noted that before maturity, a red node does not necessarily imply a zero option value. Even when  $S_t < K$ , the option retains a value linked to the possibility that the underlying will subsequently exceed the strike before maturity. Thus, only the terminal nodes satisfying  $S_T < K$  have an effectively zero value. For example, in the tree shown, certain red nodes still possess a positive value because future trajectories exist for which the option ends in-the-money.

This distinction is particularly important in the case of American options. For an American call without dividends, it is never optimal to exercise before maturity, even when the option is in-the-money. In contrast, for an American put, certain nodes may belong to the optimal exercise region, as we shall see in the following algorithms.

Once the price tree and the value matrix have been computed, the next step consists in identifying the regions where early exercise is optimal. The function `snell_envelope` serves to construct and return the Boolean matrix associated with the nodes at which immediate exercise is advantageous, materialising the set  $\mathcal{E}$  defined by (4.8). A node  $(j, i)$  is validated if the value of the Snell envelope coincides with the immediate payoff, provided that the latter is strictly positive.

```

1 def snell_envelope(M, K, r, dt, pi, N, option_type):
2     V_0, V = option_value2(M, K, r, dt, pi, N, option_type, american=
3         True)
4     exercise = np.full(np.shape(M), False)
5     for i in range(N+1):
6         for j in range(i+1):
7             immediate_gain = payoff(M[j, i], K, option_type)
8             if np.isclose(V[j, i], immediate_gain) and immediate_gain >
9                 0:
10                 exercise[j, i] = True
11     return exercise

```

As we already discussed in Chapter 4, the condition `immediate_gain > 0` is necessary. Without this second filter, a node where both the payoff and the continuation value are simultaneously zero would be categorised as an optimal exercise point, even though the gain is zero!

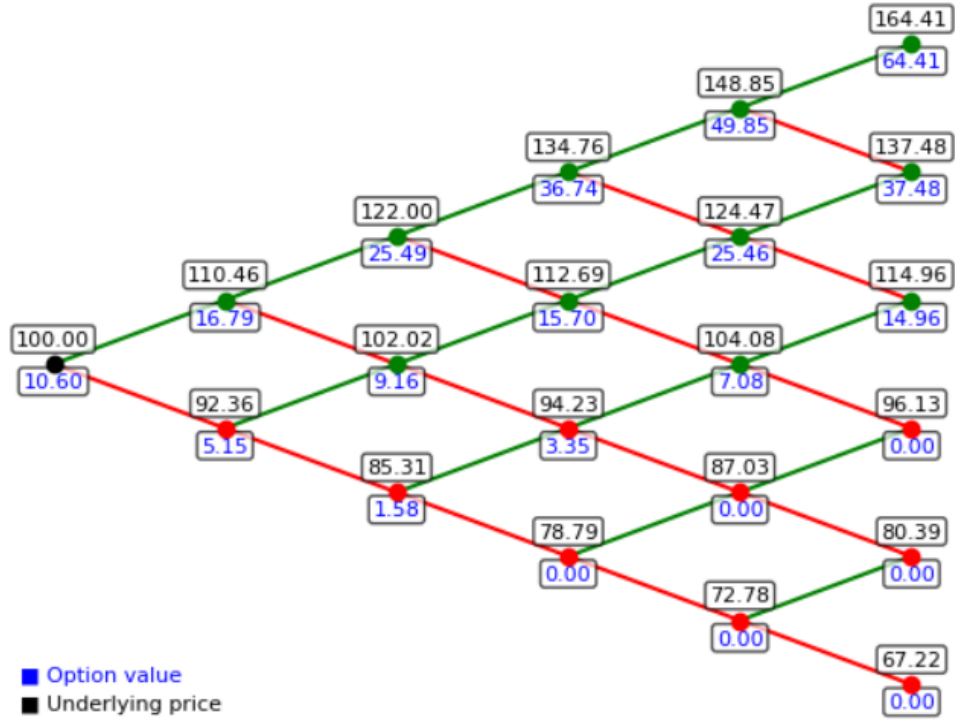


Figure 6.1: Visualisation of the binomial tree in the discrete model for a call

From this Boolean matrix, it becomes possible to isolate the trajectory separating the continuation region from the early exercise region. This is the role of the function `exercise_boundary`, which isolates for each date  $i$  the row indices  $j$  that are `True` in order to determine the critical threshold price:

```

1 def exercise_boundary(M, exercise, N, option_type):
2     time_steps = []
3     boundary_prices = []
4     for i in range(N+1):
5         exercise_indices = np.where(exercise[:,i+1] == True)[0]
6         if len(exercise_indices) > 0:
7             time_steps.append(i)
8             if option_type == 'put':
9                 boundary = np.max(exercise_indices)
10            else:
11                boundary = np.min(exercise_indices)
12            boundary_prices.append(M[boundary, i])
13     return time_steps, boundary_prices

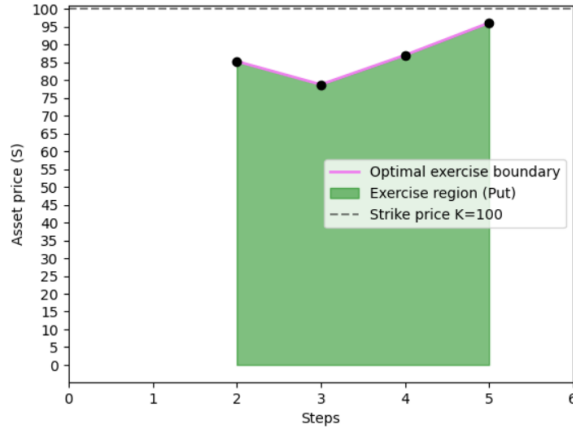
```

Of course, the selection of the limiting index depends on the nature of the option. For an American put, the holder exercises when the price of the underlying is sufficiently low (below the strike price). The algorithm therefore retains the maximal index, that is, the index corresponding to the highest possible underlying price within the exercise region, thereby materialising the upper bound of this region.

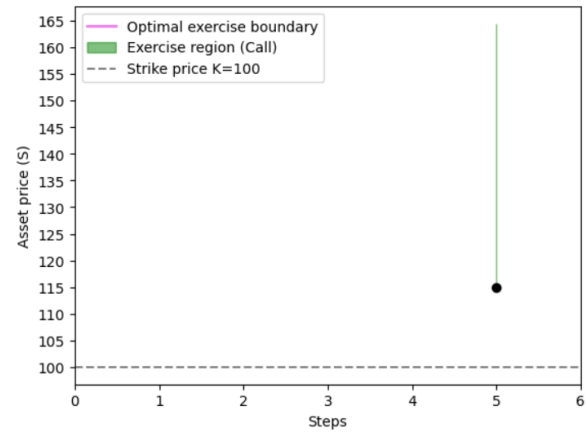
Conversely, for a call-type option, the logic is reversed and requires the identification of the lowest exercise price. It should however be recalled that in the context of a non-dividend-paying asset, by Proposition (4.3), early exercise of an American call is never optimal before maturity; this boundary will therefore remain empty, as illustrated in Figure 6.2b.

The function `visualize_snell` serves as the logical continuation of the function `exercise_boundary`. It allows the optimal exercise boundary to be plotted and the optimal exercise region to be highlighted according to the type of the option. Figure 6.2a shows two clearly distinct zones. Above the optimal exercise boundary, the white zone represents the continuation region, meaning the stock price  $S$  is too high for exercise to be advantageous. Conversely, the green zone at the bottom represents the exercise region. The stock price has then fallen below the optimal exercise boundary, shown here in pink.

We can observe that the boundary initially descends slightly before rising as maturity approaches. Indeed, when the expiry is still far off, the investor prefers to wait since the stock still has plenty of time to move and potentially fall further. This hope of realising a greater gain in the future keeps the boundary at a relatively low level. However, as maturity approaches, the investor no longer has an interest in waiting, since the stock price has less and less time to undergo further significant declines. The benefit associated with waiting then fades in favour of immediate exercise. This is why the optimal exercise boundary begins to rise, converging to the strike price  $K$  at maturity.



(a) Graphical representation of the optimal exercise region and boundary (Put)



(b) Graphical representation of the optimal exercise region and boundary (Call)

Figure 6.2: Comparison of the optimal exercise region and boundary between a Put and a Call

In order to validate the consistency of the algorithms developed, it seems essential to verify whether put-call parity is satisfied. The function `test_parity` implements this verification via equation (5.18):

$$C - P = S - Ke^{-rT}$$

```

1 def test_parity(S, K, r, T, h, b, pi, N):
2     C = option_value1(S, K, r, T, h, b, pi, N, "call")
3     P = option_value1(S, K, r, T, h, b, pi, N, "put")
4     left_side = C - P
5     right_side = S - K * np.exp(-r*T)
6     print(f"C - P = {left_side:.4f}")
7     print(f"S - Ke^(-rT) = {right_side:.4f}")
8     if np.isclose(left_side, right_side):
9         return "Put-call parity is satisfied"
10    else:
11        return "Put-call parity is not satisfied"

```

Applying the reference parameters, running the script displays the equality of both components to four decimal places:

```
C - P = 4.8771
S - Ke(-rT) = 4.8771
Put-call parity is satisfied
```

This result confirms that put-call parity is indeed satisfied on this example, which validates the consistency of the computations for these parameters.

The final step consists in confronting the results of the discrete model with the theoretical values of the continuous-time model, as we already mentioned in Chapter 5. The function `black_scholes` translates the expression of the Black-Scholes formula (cf. (5.17)):

```
1 def black_scholes(S, K, r, T, sigma):
2     d = (np.log(S/K) + r*T) / (sigma * np.sqrt(T)) + sigma * np.sqrt(T)
      / 2
3     C = S * norm.cdf(d) - K * np.exp(-r*T) * norm.cdf(d - sigma * np.
      sqrt(T))
4     return C
```

For the reference parameters, `black_scholes(100, 100, 0.05, 1, 0.2)` produces a theoretical price of approximately 10.4505. This value establishes the limit towards which the prices obtained by the binomial model converge.

In order to observe this convergence phenomenon visually as the number of steps increases, we make use of the function `convergence`<sup>2</sup>. This function computes the binomial price for different values of  $N$  and produces the comparative graph shown in Figure (5.1), relative to the theoretical Black-Scholes limiting price.

---

<sup>2</sup>The complete implementation of this function as well as the detailed theoretical analysis of the convergence and even-odd oscillations were presented in Chapter 5.

# Chapter 7

## Application: Crédit Agricole warrant

The theory of option pricing rests on an essential parameter that the model does not determine by itself: the volatility  $\sigma$  of the underlying asset. In practice, there are two approaches for estimating this quantity. The first consists in computing a *historical volatility* from past returns observed on the market. The second relies on *implied volatility*, obtained by inverting the Black-Scholes formula. We will focus solely on the first method.

The purpose of this final chapter is to confront the theory with reality. We will first estimate the historical volatility of Crédit Agricole SA stock over a period of 63 trading sessions. We will then inject this measure into our algorithm in order to price a real derivative product: a European call warrant issued by Société Générale. Finally, we will analyse the discrepancies observed between our theoretical price and the actual market quotations.

The data studied correspond to the daily closing prices of Crédit Agricole SA stock (ISIN<sup>1</sup> code: FR0000045072) over the period from 18 January 2026 to 17 April 2026, representing 63 trading sessions.

The data are imported from a CSV file and preprocessed using the following script:

```
f = pd.read_csv('Actions_Crédit_agricole_18.01-17.04.csv')
f['Date'] = pd.to_datetime(f['Date'], dayfirst=True)
f = f.sort_values('Date')
f['Fermeture'] = f['Fermeture'].str.replace(' ', '').str.replace(',', '.', '')
f['Fermeture'] = pd.to_numeric(f['Fermeture'])
```

The dates are converted to a time format (“true Python dates”) and the observations are ordered chronologically. The closing prices are then converted to numerical values after removing spaces and replacing decimal commas with decimal points. This step ensures the consistency of the return and volatility computations carried out subsequently.

The estimation of the historical volatility of Crédit Agricole stock is based on the daily log-returns of the asset.

The choice of returns rather than prices is justified by several reasons. First, returns measure the relative variations in the price and thereby allow comparisons across periods in which the stock price differs. For example, if a stock is worth 10 euros and gains 1

---

<sup>1</sup>The ISIN code (International Securities Identification Number) is a unique international identifier consisting of 12 characters. It is used to identify a financial asset (stock, bond, warrant, etc.) on global markets.

euro, the stock rises by 10%. If later the stock is worth 100 euros and gains 1 euro, it will have risen by 1%. Returns therefore allow price variations to be compared on a common basis, independently of the current stock price.

Furthermore, the prices of a stock can evolve considerably over time and are therefore not directly comparable from one period to another. Returns, by contrast, describe the relative variations in price and exhibit more regular behaviour. Finally, log-returns have the advantage of being additive: the return over several periods equals the sum of the intermediate returns. This last point greatly facilitates theoretical computations.

Log-returns are computed by comparing the closing price of each session with that of the previous day, using the following script:

```
prix = f['Fermeture']
f['Prix_Hier'] = f['Fermeture'].shift(1)
f['log_rendements'] = np.log(prix / f['Prix_Hier'])
f = f.dropna()
```

Mathematically, the log-return between two consecutive sessions is defined by

$$r_t = \ln \left( \frac{S_t}{S_{t-1}} \right)$$

Or equivalently,  $r_t = \ln \left( \frac{\text{Price}_{\text{today}}}{\text{Price}_{\text{yesterday}}} \right)$ .

The command `shift(1)` shifts the observations down by one row in order to associate each date with its previous-day value. Since the very first observation in the sample has no predecessor, it generates a missing value (NaN), which is removed using the `dropna()` method.

Once the list of log-returns has been obtained, the daily volatility is estimated by the corrected standard deviation of these variations:

```
sigma_jour = f['log_rendements'].std()
```

This yields the following unbiased estimator:

$$\hat{\sigma}_{\text{jour}} = \sqrt{\frac{1}{n-1} \sum_{t=1}^n (r_t - \bar{r})^2},$$

where  $\bar{r} = \frac{1}{n} \sum_{t=1}^n r_t$  denotes the mean of the observed log-returns.

This daily measure must then be annualised to conform to market standards<sup>2</sup>. To do so, we assume that daily returns are pairwise independent and that volatility is the same each day. Thus:

$$\hat{\sigma}_{\text{annuel}} = \hat{\sigma}_{\text{jour}} \sqrt{252}$$

where the factor 252 represents the average number of trading days on the stock exchange in a calendar year. In the implementation, this corresponds to:

```
sigma_annuel = sigma_jour * np.sqrt(252)
```

---

<sup>2</sup>Volatility is invariably quoted on an annual basis.

The numerical application yields:

$$\hat{\sigma}_{\text{annuel}} \approx 28.02\%.$$

For comparison, the implied volatility published by Société Générale for the warrant under study is 25.6%. The observed discrepancy of approximately 2.4 percentage points is not particularly surprising, as the two measures do not rest on the same approach. Indeed, historical volatility is computed from past returns, whereas implied volatility is based on market expectations regarding the future volatility of the underlying.

The two values nonetheless remain relatively close, which suggests that the estimate  $\hat{\sigma}_{\text{annuel}}$  is consistent with the market over the period studied.

In order to put the model developed previously into practice, we apply the binomial model to a European call warrant issued by Société Générale on Crédit Agricole SA stock. The main characteristics of the derivative product as well as the market parameters used are summarised in Table 7.1.

| Parameter                                                | Value             |
|----------------------------------------------------------|-------------------|
| Initial price of the underlying ( $S$ )                  | 17.78 €           |
| Strike price ( $K$ )                                     | 18.00 €           |
| Maturity ( $T$ )                                         | 0.171 year        |
| Risk-free rate ( $r$ )                                   | 2.238%            |
| Historical volatility ( $\hat{\sigma}_{\text{annuel}}$ ) | 28.02%            |
| Number of steps ( $N$ )                                  | 1000              |
| Parity (Exercise ratio)                                  | 2                 |
| Market price range (on 18/04)                            | 0.140 € – 0.150 € |

Table 7.1: Market parameters of the Crédit Agricole warrant under study

A few clarifications regarding the parameters are necessary. The maturity of the warrant is expressed in years, in accordance with the conventions used. Between the observation date (17 April 2026) and the expiry date (19 June 2026), 43 trading days elapse. Annualised on a basis of 252 working days, we obtain:

$$T = \frac{43}{252} \approx 0.171 \text{ year}$$

Note that we could have used the calendar year, since there are 63 calendar days between 17 April and 19 June, giving  $63/365 \approx 0.172$  year. This conversion ensures the consistency of the computations with the interest rate and the volatility, both of which are expressed on an annual basis.

Furthermore, the risk-free rate used is the 3-month Euribor observed at the date of the study, namely 2.238%. The Euribor (*Euro Interbank Offered Rate*) corresponds to the average interest rate at which the major banks of the euro area lend funds to one another. More precisely, the “3-month” version refers to the duration of these interbank loans. Strictly speaking, this rate is not risk-free. For example, if the borrowing bank goes bankrupt during the 3 months of the loan, the lending bank loses its money. Nevertheless, since such events are extremely rare<sup>3</sup>, this rate can be used as an approximation of the

<sup>3</sup>But not impossible, as illustrated by the bankruptcy of *Lehman Brothers* in 2008.



risk-free rate for short-maturity products. This choice is therefore consistent with the life of the warrant under study, of approximately two months (stock exchange trading days), and provides a reasonable approximation of the risk-free rate used in the binomial model.

Finally, a technical specificity of this product must be noted: its parity is equal to 2. Concretely, this means that two warrants must be held in order to exercise a right over a single Crédit Agricole share. In other words, each warrant represents only half a share. However, the binomial model computes the value of a call on a whole share. It is therefore necessary to divide the result obtained by two in order to obtain the theoretical price of one warrant.

To carry out the pricing of the warrant, we inject the parameters from Table 7.1 into our algorithm using the following script:

```
S, K, T, r, sigma, N = 17.78, 18, 0.171, 0.02238, sigma_annuel, 1000

dt, h, b, pi = paramètres(T, r, sigma, N)
M = arbre_prix(S, h, b, N)

prix, V = valeur_option2(M, K, r, dt, pi, N, "call")

print(prix / 2)
```

The binomial model then yields the following theoretical price:

$$C_{\text{warrant}} = \frac{C_{\text{binomial}}}{2} \approx 0.3757 \text{ €}$$

A significant discrepancy is observed between the theoretical price obtained, 0.3757 €, and the price observed on the market, which lies between 0.140 and 0.150 euro. Two factors may contribute to this difference. The volatility used in the model is a historical volatility (28.02%), computed over a relatively short past horizon, which overestimates the truly anticipated future volatility (recall that the implied volatility quoted by the issuer was 25.6%). However, this precision error is insufficient to explain the magnitude of the observed discrepancy.

The second and most decisive factor is the payment of a dividend of 1.13€ per share scheduled for 26 May 2026, that is, before the expiry of the warrant. Indeed, the model used in this study assumes that the underlying pays no dividend. Yet upon the payment of the dividend, the stock price will naturally fall, since the value of the dividend has left the company. This anticipated decrease in the price of the underlying reduces the probability that the call expires in-the-money and therefore diminishes its value. Since the dividend represents more than 6% of the stock price at the time of the study, its omission from our algorithm constitutes the primary source of overvaluation of the theoretical price.

This comparison highlights a limitation of pricing models: the accuracy of the result depends on the relevance of the initial assumptions. In the present case, the incorporation of dividends is indispensable for bringing the theory closer to market reality.

This dissertation has traced an arc from the most elementary foundations of arbitrage theory to a numerical application on real market data, maintaining at each step the dual guiding thread of mathematical rigour and economic meaning.

The structure adopted reflects a logical progression. The one-period model, simple as it may appear, contains the entirety of the theory: the Black-Scholes formula cannot be grasped without first having assimilated the concept of the risk-neutral measure, which is itself inseparable from the absence of arbitrage opportunities. The first fundamental theorem appears first in an intuitive form ( $S_b < (1+r)S < S_h$ ), then in an algebraic form ( $S = V^\top \beta$ ), before being transposed to the dynamic framework where the absence of arbitrage is equivalent to the existence of a probability  $\mathbb{Q}$  under which discounted prices are martingales.

Beyond the European case, we devoted a major development to American options, introducing the concept of the optimal stopping time. Through the study of the Snell envelope, we showed that the valuation of these assets no longer rests on a classical martingale dynamic, but requires the introduction of a supermartingale structure. The implementation of the dynamic programming algorithm via backward induction made it possible to formalise the optimal exercise boundary numerically. This theoretical detour shed light on the duality between financial valuation and stochastic optimisation problems.

The Black-Scholes formula was obtained without postulating continuous time, but as the limit of the Cox, Ross and Rubinstein binomial model as  $N \rightarrow +\infty$ , by virtue of the Lindeberg theorem. This allows a continuous-time formula to be recovered from a discrete model that is entirely under our control. Finally, confronting our script with the real prices of the Cr dit Agricole SA warrant served as a reminder that a model remains an approximation. The observed discrepancy is explained primarily by a limitation of our implementation, namely the failure to account for dividends paid by the stock.

This work may be extended in two directions. The first is mathematical, involving the direct passage to continuous time via Brownian motion and It  stochastic calculus in order to recover the Black-Scholes formula without going through the discrete model. The second is more practical, encompassing the integration of dividends into the tree and the study of stochastic volatility (the Heston model) or local volatility (the Dupire model).

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